

2009-10

Academic Regulations 2009 for B. Tech (Regular)

(Effective for the students admitted into I year
from the Academic Year 2009-2010 onwards)

1. Award of B.Tech. Degree

A student will be declared eligible for the award of the B.Tech. Degree if he fulfils the following academic regulations:

- i. Pursue a course of study for not less than four academic years and in not more than eight academic years.
 - ii. Register for 220 credits and secure all 220credits
2. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech course and their admission is cancelled.

3. Courses of study

The courses of study are offered at present for specialization for the B. Tech. Course:

S.No.	Branch
1.	Aeronautical Engineering.
2.	Biotechnology.
3.	Civil Engineering.
4.	Computer Science and Engineering.
5.	Computer Science and System Engineering.
6.	Electrical and Electronics Engineering.
7.	Electronics and Communication Engineering.
8.	Electronics and Computer Engineering.
9.	Electronics and Control Engineering.
10.	Electronics and Instrumentation Engineering.
11.	Information Technology.
12.	Mechanical Engineering.

and any other course as approved by the authorities of the University from time to time.

2009-10**4. Credits**

	I Year		Semester	
	Periods / Week	Credits	Periods / Week	Credits
Theory	03	06	03	04
	02	04	--	--
Practical	03	04	03	02
Drawing	06	06	03	02
			06	04
Seminar	--	--	6	02
Project	--	--	15	10

5. Distribution and Weightage of Marks

- The performance of a student in each semester / I year shall be evaluated subject –wise with a maximum of 100 marks for theory and 75 marks for practical subject. In addition seminar and project work shall be evaluated for 50 and 200 marks respectively.
- For theory subjects the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End-Examination.
- For theory subjects, during the semester there shall be Two midterm examinations. Each mid term examination consists of objective paper for 10 marks and subjective paper for 20 marks with duration of 1hour 50 minutes (20 minutes for objective and 90 minutes for subjective paper).

Objective paper is set for 20 bits for 10 marks. Subjective paper shall contain 5 questions of which student has to answer 3 questions evaluated* for 20 marks. First mid term examination shall be conducted for I-IV units of syllabus and second mid term examination shall be conducted for V -VIII units. The total marks secured by the student in each mid term examination for 30 marks is considered and the better of the two mid term examinations shall be taken as the final sessional marks secured by each candidate in the subject.

However for first year, there shall be Three midterm examinations as in the above pattern and the average marks of the best two midterm examinations secured in each subject shall be considered as final marks for sessionals.

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*Note 1: The subjective paper shall contain 5 questions of equal weightage of 10 marks and the marks obtained for 3 questions shall be condensed to 20 marks, any fraction rounded off to the next higher mark

*Note 2: The mid term examination shall be conducted first by distribution of the Objective paper simultaneously marking the attendance, after 20 minutes the answered objective paper is collected back. The student is not allowed to leave the examination hall. Then the descriptive question paper and the answer booklet are distributed. After 90 minutes the answered booklets are collected back.

- iv. For practical subjects there shall be a continuous evaluation during the semester for 25 sessional marks and 50 end examination marks. Day-to-day work in the laboratory shall be evaluated for 25 marks by the concerned laboratory teacher based on the report of experiments/jobs. The end examination shall be conducted by the laboratory teacher and another examiner.
- v. For the subject having design and / or drawing, such as Engineering Drawing, Machine Drawing and estimation, the distribution shall be 30 marks for internal evaluation and 70 marks for end examination. The Internal evaluation for sessionals will be 15 marks for day-to-day work in the class that shall be evaluated by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two midterm exams in a Semester for a duration of 2hrs each, evenly distributed over the syllabi for 15 marks and the better of the two shall be considered as internal test marks. The sum of day to day evaluation and the internal test marks will be the final sessionals for the subject. However in the I year class, there shall be three midterm exams and the average of best two will be taken into consideration.
- vi. There shall be a seminar presentation in IV year II Semester. For the seminar, the student shall collect the information on a specialized topic and prepare a technical report, showing his understanding over the topic, and submit to the department before presentation. The report and the presentation shall be evaluated by the Departmental committee consisting of Head of the department, seminar supervisor and a senior faculty member. The

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- seminar shall be evaluated for 50 marks and marks shall be submitted to the University along with internal marks. There shall be no external examination for seminar.
- vii. Out of a total of 200 marks for the project work, 60 marks shall be for Internal Evaluation and 140 marks for the End Semester Examination (Viva-voce). The viva-voce shall be conducted by a committee consisting of HOD, Project Supervisor and an External Examiner nominated by the University. The evaluation of project work shall be conducted at the end of the IV year. The Internal Evaluation shall be made by the departmental committee, on the basis of two seminars given by each student on the topic of his project.
 - viii. Laboratory marks and the sessional marks awarded by the College are not final. They are subject to scrutiny and scaling by the University wherever necessary. In such cases, the sessional and laboratory marks awarded by the College will be referred to a Committee. The Committee will arrive at a scaling factor and the marks will be scaled as per the scaling factor. The recommendations of the Committee are final and binding.
 - ix. The laboratory records and internal test papers shall be preserved in the respective institutions as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

6. Attendance Requirements:

- i. A student shall be eligible to appear for University examinations if he acquires a minimum of 75% of attendance in aggregate of all the subjects in a semester/ I year.
- ii. **Shortage of Attendance below 65% in aggregate shall in NO case be condoned.**
- iii. Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester or I year may be granted by the College Academic Committee.
- iv. Students whose shortage of attendance is not condoned in any semester / I year are not eligible to take their end examination of that class and their registration shall stand cancelled.
- v. A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester / I

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year, as applicable. They may seek readmission for that semester / I year when offered next.

- vi. A stipulated fee shall be payable towards condonation of shortage of attendance to the University.

7. Minimum Academic Requirements:

The following academic requirements have to be satisfied in addition to the attendance requirements mentioned in item no.6

- i. A student shall be deemed to have satisfied the minimum academic requirements and earned the credits allotted to each theory, practical, design, drawing subject or project if he secures not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the internal evaluation and end examination taken together. In the Seminar he should secure 40%.
 - ii. A student shall be promoted from II to III year only if he fulfils the academic requirement of securing **40** credits from
 - a. One regular and one supplementary examinations of I year.
 - b. One regular examination of II year I semester irrespective of whether the candidate takes the end examination or not as per the normal course of study.
 - iii. A student shall be promoted from third year to fourth year only if he fulfils the academic requirements of securing **68** credits from the following examinations,
 - a. Two regular and two supplementary examinations of I year.
 - b. Two regular and one supplementary examinations of II year I semester.
 - c. One regular and one supplementary examinations of II year II semester.
 - d. One regular examination of III year I semester. irrespective of whether the candidate takes the end examination or not as per the normal course of study.

And in case of getting detained for want of credits by sections ii and iii above, the student may make up the credits through supplementary exams of the above exams before the date of class work commencement of Third or Fourth year I semester respectively.

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- iv. A student shall register and put up minimum attendance in all 220 credits and earn all the 220 credits. Marks obtained in all 220 credits shall be considered for the calculation of percentage of marks obtained.
 - v. Students who fail to earn 220 credits as indicated in the course structure within eight academic years from the year of their admission shall forfeit their seat in B.Tech course and their admission shall stand cancelled.
- 8. Course pattern:**
- i. The entire course of study is of four academic years. The first year shall be on yearly pattern and the second, third and fourth years on semester pattern.
 - ii. A student eligible to appear for the end examination in a subject, but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.
 - iii. When a student is detained due to lack of credits / shortage of attendance he may be re-admitted when the semester is offered after fulfilment of academic regulations, whereas he continues to be in the academic regulations he was first admitted.
- 9. Transitory Regulations:**
- Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of class work with the same or equivalent subjects as and when subjects are offered, subject to Section 2. and they continues to be in the academic regulations they were first admitted.
- 10. With-holding of results:**
- If the candidate has any dues not paid to the university or if any case of indiscipline or malpractice is pending against him, the result of the candidate shall be withheld and he will not be allowed / promoted into the next higher semester. The issue of degree is liable to be withheld in such cases.

2009-10**11. Award of Class:**

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes:

Class Awarded	% of marks to be secured	From the aggregate marks secured for the best 220 Credits.
First Class with Distinction	70% and above	
First Class	Below 70% but not less than 60%	
Second Class	Below 60% but not less than 50%	
Pass Class	Below 50% but not less than 40%	

(The marks in internal evaluation and end examination shall be shown separately in the marks memorandum)

12. Minimum Instruction Days:

The minimum instruction days including exams for each semester / I year shall be 90/180 days respectively.

13. There shall be no branch transfers after the completion of admission process.

14. There shall be no place transfer within the Constituent Colleges.

15. General:

- i. The academic regulations should be read as a whole for purpose of any interpretation.
- ii. Malpractices rules- nature and punishments is appended
- iii. Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.
- iv. In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.
- v. The University may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on roles with effect from the dates notified by the University.

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ACADEMIC REGULATIONS FOR B. TECH. (LATERAL ENTRY SCHEME)

(Effective for the students getting admitted into II year through Lateral Entry Scheme from the Academic Year 2010-2011 and onwards)

1. **Award of B.Tech. Degree**

A student admitted in LES will be declared eligible for the award of the B. Tech Degree if he fulfils the following academic regulations:

- i. Pursue a course of study for not less than three academic years and in not more than six academic years.
- ii. Register for 168 credits and secure all 168 credits from II to IV year of Regular B.Tech. program
2. Students, who fail to fulfil the requirement for the award of the degree in six consecutive academic years from the year of admission, shall forfeit their seat.
3. The regulations **3** to **6** are to be adopted as that of B. Tech. (Regular).

7. **Minimum Academic Requirements :**

The following academic requirements have to be satisfied in addition to the attendance requirements mentioned in item no.6

- i. A student shall be deemed to have satisfied the minimum academic requirements and earned the credits allotted to each theory, practical, design, drawing subject or project if he secures not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the internal evaluation and end examination taken together. For the Seminar he should secure 40% in the internal evaluation.
- ii. A student shall be promoted from third year to fourth year only if he fulfils the academic requirements of 42 credits from the following examinations.
 - a. Two regular and one supplementary examinations of II year I semester.
 - b. One regular and one supplementary examinations of II year II semester.
 - c. One regular examination of III year I semester.
irrespective of whether the candidate takes the end examination or not as per the normal course of study.
and in case of getting detained for want of credits the student may make up the credits through supplementary exams of the above

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exams before the date of class work commencement of Fourth year I semester.

8. Course Pattern

- i. The entire course of study is three academic years on semester pattern.
 - ii. A student eligible to appear for the end examination in a subject, but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.
 - iii. When a student is detained due to lack of credits / shortage of attendance he may be re-admitted when the semester is offered after fulfilment of academic regulations, whereas he continues to be in the academic regulations he was first admitted.
9. The regulations 9 to 10 are to be adopted as that of B. Tech. (Regular).

11. Award of Class:

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes:

First Class with Distinction	70% and above	From the aggregate marks secured for 168 Credits. (i.e. II year to IV year)
First Class	Below 70% but not less than 60%	
Second Class	Below 60% but not less than 50%	
Pass Class	Below 50% but not less than 40%	

(The marks in internal evaluation and end examination shall be shown separately in the marks memorandum)

12. The regulations 12 to 15 are to be adopted as that of B. Tech. (Regular). All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme)

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**RULES FOR
DISCIPLINARY ACTION FOR MALPRACTICES / IMPROPER
CONDUCT IN EXAMINATIONS**

	Nature of Malpractices/Improper conduct	Punishment
	<i>If the candidate:</i>	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including

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	practical) in which the candidate is appearing.	practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the University.
3.	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.
4.	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question	Expulsion from the examination hall and cancellation of performance in

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	paper during the examination or answer book or additional sheet, during or after the examination.	that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the Chief Superintendent/Assistant – Superintendent / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation,	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police

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	assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8.	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the

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		performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat.
9.	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and

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		project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester/year examinations.
12.	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the University for further action to award suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the candidates as per the above guidelines.
2. Punishment for institutions : (if the squad reports that the college is also involved in encouraging malpractices)
 - (i) A show cause notice shall be issued to the college.
 - (ii) Impose a suitable fine on the college.

Shifting the examination centre from the college to another college for a specific period of not less than one year.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

III B. Tech. – I Semester (E.C.E)

S. No.	Group	Subject	L	T	P	CP
01.	9A02503	Control Systems	3	1	0	4
02.	9A04501	Analog Communications	3	1	0	4
03.	9A04502	Linear IC Applications	3	1	0	4
04.	9A04503	Antennas & Wave Propagation	3	1	0	4
05.	9A05406	Computer Organization	4	0	0	4
06.	9A04504	Digital IC Applications	3	1	0	4
07.	9A04505	Linear & Digital IC Applications Lab	0	0	3	2
08.	9A04506	Pulse & Digital Circuits Lab	0	0	3	2
		contact periods / week	19	05	06	
			Total/Week 30			
Total Credits (6 Theory + 2 Labs)						28

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

III B. Tech. – II Semester (E.C.E)

S. No.	Group	Subject	L	T	P	CP
01.	9A04601	Digital Communications	3	1	0	4
02.	9A04602	Microprocessors & Microcontrollers	4	0	0	4
03.	9A04603	Digital Signal Processing	3	1	0	4
04.	9A04604	Electronic Measurements and Instrumentation	4	0	0	4
05.	9A04605	VLSI Design	4	0	0	4
06.	9A04606	Microwave Engineering	3	1	0	4
07.	9A04607	Analog & Digital Communications lab	0	0	3	2
08.	9AHS601	Advanced English Language Communication skills lab	0	0	3	2
		contact periods / week	21	03	06	
			Total/Week 30			
Total Credits (6 Theory + 2 Labs)						28

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**JAWAHARLAL NEHRU
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IV B. Tech. – I Semester (E.C.E)

S. No.	Group	Subject	L	T	P	CP
01.	9AHS701	Management Science	4	0	0	4
02.	9A04701	Embedded Real Time Operating Systems	3	1	0	4
03.	9A05506	Computer Networks	4	0	0	4
04.	9A04702	Optical Communications	3	1	0	4
05.	9A04703 9A04704 9A05505	Elective – I 1. RADAR Systems 2. DSP Processors & Architectures 3. Operating Systems	3	1	0	4
06.	9A10802 9A04705 9A04706	Elective – II 1. Bio-Medical Instrumentation 2. T.V. Engineering 3. Digital Design through Verilog HDL	4	0	0	4
07.	9A04707	Microwave & Optical Communications Lab	0	0	3	2
08.	9A04708	Microprocessors & DSP lab	0	0	3	2
		contact periods / week	21	03	06	
			Total/Week 30			
Total Credits (6 Theory + 2 Labs)						28

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**JAWAHARLAL NEHRU
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IV B.Tech. – II Semester (E.C.E)

S. No.	Group	Subject	L	T	P	C P
01.	9A04801	Cellular & Mobile Communications	3	1	0	4
02.	9A04802	Digital Image Processing	3	1	0	4
03.	9A04803 9A04804 9A05709	Elective – III 1. Satellite Communications 2. Spread Spectrum Communications 3. Information Security	3	1	0	4
04.	9A04805 9A04806 9A04807	Elective – IV 1. Adaptive Filter Theory 2.Wireless Communications & Networks 3. Data Communications	3	1	0	4
05.	9A04808	Seminar	0	0	0	2
06.	9A04809	Project	--	--	--	10
		contact periods / week	12	04	0	
			Total/Week 16			
Total Credits (4Theory + Seminar + Project Work)						28

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**JAWAHARLAL NEHRU
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Electronics and Communication Engineering

(9A02503) CONTROL SYSTEMS

(Common to EEE, ECE, E Con E, EIE)

B.Tech III-I Sem. (E.C.E.)

T	P	C
4	0	4

Objective:

In this course it is aimed to introduce to the students the principles and applications of control systems in everyday life. The basic concepts of block diagram reduction, time domain analysis solutions to time invariant systems and also deals with the different aspects of stability analysis of systems in frequency domain and time domain.

UNIT I

INTRODUCTION

Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Examples of control systems- Classification of control systems, Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations, Impulse Response and transfer functions - Translational and Rotational mechanical systems.

UNIT II

TRANSFER FUNCTION REPRESENTATION

Transfer Function of DC Servo motor - AC Servo motor- Synchro transmitter and Receiver -Block diagram algebra –Signal flow graph - Reduction using Mason's gain formula.

UNIT III

TIME RESPONSE ANALYSIS

Standard test signals - Time response of first order systems – Characteristic Equation of Feedback control systems, Transient response of second order systems - Time domain specifications – Steady state response - Steady state errors and error constants – Effects of proportional, integral, derivative Controls.

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UNIT IV**STABILITY ANALYSIS IN S-DOMAIN**

The concept of stability – Routh's stability criterion – qualitative stability and conditional stability – limitations of Routh's stability. The root locus concept - construction of root loci-effects of adding poles and zeros to $G(s)H(s)$ on the root loci.

UNIT V**FREQUENCY RESPONSE ANALYSIS**

Introduction, Frequency domain specifications-Bode diagrams-Determination of Frequency domain specifications and transfer function from the Bode Diagram-Phase margin and Gain margin-Stability Analysis from Bode Plots.

UNIT VI**STABILITY ANALYSIS IN FREQUENCY DOMAIN**

Polar Plots-Nyquist Plots-Stability Analysis.

UNIT VII**CLASSICAL CONTROL DESIGN TECHNIQUES**

Compensation techniques – Lag, Lead, Lead-Lag Controllers design in frequency Domain, P, PD, PI, PID Controllers.

UNIT VIII**STATE SPACE ANALYSIS OF CONTINUOUS SYSTEMS**

Concepts of state, state variables and state model, derivation of state models from block diagrams, Diagonalization- Solving the Time invariant state Equations- State Transition Matrix and its Properties.

TEXT BOOKS:

1. Automatic Control Systems 8th edition– by B. C. Kuo 2003– John wiley and son's.
2. Control Systems Engineering – by I. J. Nagrath and M. Gopal, New Age International (P) Limited, Publishers, 5th edition, 2007.

REFERENCES:

1. Modern Control Engineering – by Katsuhiko Ogata – Prentice Hall of India Pvt. Ltd., 5th edition, 2010.
2. Control Systems by N.K.Sinha, New Age International (P) Limited Publishers, 3rd Edition, 1998.
3. Control Systems Engg. by NISE 3rd Edition – John wiley.
4. “Modelling & Control Of Dynamic Systems” by Narciso F. Macia George J. Thaler, Thomson Publishers.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04501) ANALOG COMMUNICATIONS**

B.Tech III-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**INTRODUCTION:**

Elements of communication Systems - information, Messages and Signals, Fundamental Limitations of communication Systems, Modulation & Coding–Modulation Methods, Modulation Benefits and Applications, Overview of Coding Methods and Benefits.

UNIT II**LINEAR CONTINUOUS WAVE (CW) MODULATION - I:**

Band Pass Signals and Systems – Analog Message Conventions, Band Pass Signals, Band Pass Transmission, Bandwidth, Double Side Band Amplitude Modulation – AM Signals and Spectra, DSB Signals and Spectra, Tone Modulation and Phaser Analysis, Modulators and Transmitters – Product Modulators, Square Law Modulators, Balanced Modulators and Switching Modulators.

UNIT III**LINEAR CONTINUOUS WAVE (CW) MODULATION – II:**

Suppressed Side Band Amplitude Modulation - Single Side Band Signals and Spectra, Single Side Band Generation, Vestigial Side Band Signals and Spectra, Frequency Conversion and Demodulation – Frequency Conversion, Synchronous Detection, Envelope Detection, Illustrative Problems.

UNIT IV**ANGLE CONTINUOUS WAVE (CW) MODULATION - I:**

Phase and Frequency Modulation – PM and FM Signals, Narrow Band PM and FM, Tone Modulation, Multi Tone and Periodic Modulation, Transmission Bandwidth and Distortion – Transmission Estimates, Linear Distortion, Non-Linear Distortion and Limiters.

UNIT V**ANGLE CONTINUOUS WAVE (CW) MODULATION – II:**

Generation and Detection of PM and FM – Direct FM and VCOs, Phase Modulators and Indirect FM, Triangular wave FM, Frequency Detection, Interference – Interfering Sinusoids, Pre-Emphasis and De-Emphasis Filtering, FM Capture Effect, Illustrative Problems.

UNIT VI**ANALOG COMMUNICATION SYSTEMS:**

Receivers for CW Modulation – Super Heterodyne receivers, direct conversion receiver, special purpose receivers, Receiver Specifications, Receiver Measurements, Multiplexing Systems, synchronous detection and frequency synthesizers using Phase Locked Loop (PLL), Linearized PLL FM detection, Illustrative Problems.

UNIT VII**NOISE:**

Thermal Noise & Available Power, White noise and filtered noise, Noise equivalent bandwidth, base band signal Transmission with noise- Additive Noise & S/N, Analog Signal Transmission, Noise in Analog Modulation Systems- Band Pass Noise System Models, Quadrature Components, envelope phase, Correlation Functions, Linear CW Modulation with Noise – Synchronous Detection, Envelope Detection, and Threshold Effect, Angle CW with Noise, Post detection Noise, Destination S/N, FM Threshold Effect, Analog Pulse modulation with Noise, Illustrative Problems.

UNIT VIII**PULSE MODULATION TECHNIQUES:**

Pulse amplitude modulation – Flat top sampling and Pulse amplitude modulation (PAM), Pulse-Time Modulation – Pulse Duration and Pulse Position modulations, PPM spectral analysis, Illustrative Problems.

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TEXT BOOKS:

1. A. Bruce Carlson, & Paul B. Crilly, “Communication Systems – An Introduction to Signals & Noise in Electrical Communication”, McGraw-Hill International Edition, 5th Edition, 2010.
2. Sham Shanmugam, “Digital and Analog Communication Systems”, Wiley-India edition, 2006.

REFERENCES:

1. Simon Haykin, “Communication Systems”, Wiley-India edition, 3rd edition, 2010.
2. B.P. Lathi, & Zhi Ding, “Modern Digital & Analog Communication Systems”, Oxford University Press, International 4th edition, 2010.
3. Herbert Taub & Donald L Schilling, “Principles of Communication Systems”, Tata McGraw-Hill, 3rd Edition, 2009.
4. R.E. Ziemer & W.H. Tranter, “Principles of Communication-Systems Modulation & Noise”, Jaico Publishing House, 2001.
5. George Kennedy and Bernard Davis, “Electronics & Communication System”, TMH, 2004.

2009-10

**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04502) LINEAR IC APPLICATIONS**

B.Tech III-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I**INTEGRATED CIRCUITS**

Differential amplifier –DC and AC analysis of Dual input balanced output configuration, Properties of other differential amplifier configuration (dual input unbalanced output, single ended input-balanced/unbalanced output), DC coupling and cascade differential amplifier stages, Level Translator.

UNIT-II

Characteristics of OP-Amps, integrated circuits-types, classification, package types and temperature ranges, power supplies, OP-Amp Block diagram, ideal and practical OP-Amp specifications, DC and AC characteristics, 741 OP-Amp and its features, FET input OP-Amps, OP-Amp parameters and measurement, input and output offset voltages and currents, slew rate, CMRR, PSRR, drift, Frequency compensation technique.

UNIT-III**LINEAR APPLICATIONS OF OP-AMPS**

Inverting and non-inverting amplifier, integrator and differentiator, difference amplifier, instrumentation amplifier, AC amplifier, V-I, I-V converters, Buffers.

UNIT-IV**NON LINEAR APPLICATIONS OF OP-AMPS**

Non-linear function generation, comparators, Multivibrators, Triangular and square wave generators, Log and antilog amplifiers, precision rectifiers.

2009-10

UNIT-V**ANALOG FILTERS**

Introduction, Butterworth filters-first order, second order LPF, HPF filters. Band pass, Band reject and all pass filters.

UNIT-VI**TIMERS AND PHASE LOCKED LOOPS**

Introduction to 555 Timer, functional diagram, Monostable and Astable operations and applications, Schmitt Trigger, PLL-Introduction, Block schematic, principles and description of individual blocks, 555 PLL, applications of PLL-Frequency multiplication, frequency translation, AM, FM and FSK demodulators.

UNIT-VII**D/A AND A/D CONVERTERS**

Introduction, Basic DAC techniques, weighted resistor DAC, R-2R Ladder DAC, Inverted R-2R DAC and IC 1408 DAC, different types of ADCs-parallel comparator type ADC, counter type ADC, successive approximation ADC and Dual slope ADC. DAC and ADC specifications, specifications AD 574 (12 bit ADC).

UNIT-VIII**ANALOG MULTIPLIERS AND MODULATORS**

Four quadrant multiplier, Balanced modulator, IC 1496, applications of analog switches and multiplexers, sample and hold amplifiers.

TEXT BOOKS:

1. Op-Amps & Linear ICs - Ramakanth A. Gayakwad, 4th edition, PHI, 1987.
2. Operational Amplifiers & Linear Integrated Circuits-R.F.Coughlin & Fredrick Driscoll, 6th edition, PHI.

REFERENCES:

1. Operational Amplifiers & Linear ICs by David A. Bell, 2nd edition, Oxford University Press, 2010.

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2. Linear Integrated Circuits – D. Roy Chowdhury, New Age International (p) Ltd, 2nd Edition, 2003.
3. Design with Operational Amplifiers & Analog Integrated Circuits - Sergio Franco, McGraw Hill, 1988.
4. Operational Amplifiers – C.G. Clayton, Butterworth & Company Publ. Ltd./ Elsevier, 1971.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04503) ANTENNAS & WAVE PROPAGATION**

B.Tech III-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**ANTENNA BASICS:**

Introduction, Basic antenna parameters- patterns, Beam Area, Radiation Intensity, Beam Efficiency, Directivity-Gain-Resolution, Antenna Apertures, Effective height, Illustrative problems. Fields from oscillating dipole, Field Zones, Shape-Impedance considerations, Antenna temperature, front-to-back ratio, antenna theorems, radiation-basic Maxwell's equations, retarded potential-Helmholtz Theorem.

UNIT II**THIN LINEAR WIRE ANTENNAS:**

Radiation from Small Electric Dipole, Quarter wave Monopole and Half wave Dipole – Current Distributions, Field Components, Radiated power, Radiation Resistance, Beam width, Directivity, Effective Area and Effective Height. Natural current distributions, far fields and patterns of Thin Linear Center-fed Antennas of different lengths, Illustrative problems. Loop Antennas: Introduction, Small Loop, Comparison of far fields of small loop and short dipole, Radiation Resistances and Directives of small and large loops (Qualitative Treatment).

UNIT III**ANTENNA ARRAYS:**

Point sources- Definition, Patterns, arrays of 2 Isotropic sources-Different cases. Principle of Pattern Multiplication, Uniform Linear Arrays – Broadside Arrays, Endfire Arrays, EFA with Increased Directivity, Derivation of their characteristics and comparison, BSAA with Non-uniform Amplitude Distributions- General considerations and Bionomial Arrays, Illustrative problems.

UNIT IV**VHF, UHF AND MICROWAVE ANTENNAS - I:**

Arrays with Parasitic Elements, Yagi - Uda Arrays, Folded Dipoles & their characteristics. Helical Antennas-Helical Geometry, Helix modes, Practical Design considerations for Monofilar Helical Antenna in Axial and Normal Modes. Horn Antennas- Types, Fermat's Principle, Optimum Horns, Design Considerations of Pyramidal Horns, Illustrative Problems.

UNIT V**VHF, UHF AND MICROWAVE ANTENNAS - II:**

Micro strip Antennas- Introduction, features, advantages and limitations, Rectangular patch antennas- Geometry and parameters, characteristics of Micro strip antennas, Impact of different parameters on characteristics, reflector antennas- Introduction, Flat sheet and corner reflectors, paraboloidal reflectors- geometry, pattern characteristics, Feed Methods, Reflector Types- Related Features, Illustrative Problems.

UNIT VI**LENS ANTENNAS:**

Introduction, Geometry of Non-metallic Dielectric Lenses, Zoning , Tolerances, Applications.

Antenna Measurements: Introduction, Concepts- Reciprocity, Near and Far Fields, Coordination system, sources of errors, Patterns to be Measured, Pattern Measurement Arrangement, Directivity Measurement , Gain Measurements (by comparison, Absolute and 3-Antenna Methods).

UNIT VII**WAVE PROPAGATION - I:**

Introduction, Definitions, Characterizations and general classifications, different modes of wave propagation, Ray/ Mode concepts. Ground wave propagation (Qualitative treatment)- Introduction, Plane earth reflections, Space and surface waves, wave tilt, curved earth reflections. Space wave propagation- Introduction, field strength variation with distance and height, effect of earth's curvature, absorption. Super

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refraction, M-curves and duct propagation, scattering phenomena, tropospheric propagation, fading and path loss calculations.

UNIT VIII**WAVE PROPAGATION – II:**

Sky wave propagation- Introduction, structure of Ionosphere, refraction and reflection of sky waves by Ionosphere, Ray path, Critical frequency, MUF, LUF, OF, Virtual height and Skip distance, Relation between MUF and Skip distance, Multi-HOP propagation, Energy loss in Ionosphere, Summary of Wave Characteristics in different frequency ranges.

TEXT BOOKS:

1. Antennas and wave propagation – John D. Kraus and Ronald J. Marhefka and Ahmad S.Khan, TMH, New Delhi, 4th Ed., (special Indian Edition), 2010
2. Electromagnetic Waves and Radiating Systems – E.C. Jordan and K.G. Balmain, PHI, 2nd ed., 2000.

REFERENCES:

1. Antenna Theory - C.A. Balanis, John Wiley & Sons, 2nd ed., 2001.
2. Antennas and Wave Propagation – K.D. Prasad, Satya Prakashan, Tech India Publications, New Delhi, 2001.
3. Transmission and Propagation – E.V.D. Glazier and H.R.L. Lamont, The Services Text Book of Radio, vol. 5, Standard Publishers Distributors, Delhi.
4. Electronic and Radio Engineering – F.E. Terman, McGraw-Hill, 4th edition, 1955.
5. Antennas – John D. Kraus, McGraw-Hill (International Edition), 2nd ed., 1988.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A05406) COMPUTER ORGANIZATION**

B.Tech III-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I**BASIC STRUCTURE OF COMPUTERS:**

Computer Types, Functional Units, Basic Operational Concepts, Bus Structures, Software, Performance, Multiprocessors and Multi Computers. Data Representation- Fixed Point Representation, Floating – Point Representation. Error Detection Codes.

UNIT II**REGISTER TRANSFER AND MICROOPERATIONS:**

Register Transfer Language. Register Transfer, Bus and Memory Transfers, Arithmetic Microoperations, Logic Microoperations, Shift Micro Operations, Arithmetic Logic Shift Unit, Instruction Codes, Computer Registers, Computer Instructions, Instruction Cycle. Memory: Reference Instructions- Input – Output and Interrupt, STACK Organization. Instruction Formats, Addressing Modes, DATA Transfer and Manipulation, Program Control, Reduced Instruction Set Computer.

UNIT III**MICRO PROGRAMMED CONTROL:**

Control Memory, Address Sequencing, Microprogram Example, Design of Control Unit Hard Wired Control, Microprogrammed Control.

UNIT IV**COMPUTER ARITHMETIC:**

Addition and Subtraction, Multiplication Algorithms, Division Algorithms, Floating – Point Arithmetic Operations, Decimal Arithmetic Unit Decimal Arithmetic Operations.

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UNIT V**THE MEMORY SYSTEM:**

Basic concepts, semiconductor RAM memories, Read-only memories, Cache memories, performance considerations, Virtual memories, secondary storage, Introduction to RAID.

UNIT VI**INPUT-OUTPUT ORGANIZATION:**

Peripheral Devices, Input-Output Interface, Asynchronous data transfer Modes of Transfer, Priority Interrupt, Direct memory Access, Input – Output Processor (IOP), Serial communication; Introduction to peripheral component, Interconnect (PCI) bus. Introduction to standard serial communication protocols like RS232, USB, IEEE1394.

UNIT VII**PIPELINE AND VECTOR PROCESSING:**

Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.

UNIT VIII**MULTI PROCESSORS:**

Characteristics of Multiprocessors, Interconnection Structures, Interprocessor Arbitration, InterProcessor Communication and Synchronization, Cache Coherence, Shared Memory Multiprocessors.

TEXT BOOKS:

1. Computer Organization – Carl Hamacher, Zvonks Vranesic, SafeaZaky, 5th Edition, McGraw Hill.
2. Computer Systems Architecture – M.Moris Mano, 3rd Edition, Pearson/PHI.

REFERENCES:

1. Computer Organization and Architecture – William Stallings 6th Edition, Pearson/PHI.
2. Structured Computer Organization – Andrew S. Tanenbaum, 4th Edition PHI/Pearson.

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3. Fundamentals of Computer Organization and Design, - Sivarama Dandamudi Springer Int. Edition.
4. Computer Architecture a quantitative approach, John L. Hennessy and David A. Patterson, 4th Edition, Elsevier.
5. Computer Architecture: Fundamentals and principles of Computer Design, Joseph D. Dumas II, BS Publication.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04504) DIGITAL IC APPLICATIONS**

B.Tech III-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I**CMOS LOGIC:**

Introduction to logic families, CMOS logic, CMOS steady state electrical behavior, CMOS dynamic electrical behavior, CMOS logic families.

UNIT II**BIPOLAR LOGIC AND INTERFACING:**

Bipolar logic, Transistor logic, TTL families, CMOS/TTL interfacing, low voltage CMOS logic and interfacing, Emitter coupled logic, Comparison of logic families, Familiarity with standard 74XX and CMOS 40XX series-ICs – Specifications.

UNIT III**THE VHDL HARDWARE DESCRIPTION LANGUAGE:**

Design flow, program structure, types and constants, functions and procedures, libraries and packages.

UNIT IV**THE VHDL DESIGN ELEMENTS:**

Structural design elements, data flow design elements, behavioral design elements, time dimension and simulation synthesis.

UNIT V**COMBINATIONAL LOGIC DESIGN:**

Decoders, encoders, three state devices, multiplexers and demultiplexers, Code Converters, EX-OR gates and parity circuits, comparators, adders & subtractors, ALUs, Combinational multipliers. VHDL modes for the above ICs.

UNIT VI**DESIGN EXAMPLES (USING VHDL):**

Design examples (using VHDL) - Barrel shifter, comparators, floating-point encoder, dual parity encoder.

UNIT VII**SEQUENTIAL LOGIC DESIGN:**

Latches and flip-flops, PLDs, counters, shift register, and their VHDL models, synchronous design methodology, impediments to synchronous design.

UNIT VIII**MEMORIES:**

ROMs: Internal structure, 2D-decoding commercial types, timing and applications.

Static RAM: Internal structure, SRAM timing, standard SRAMS, synchronous SRAMS.

Dynamic RAM: Internal structure, timing, synchronous DRAMs. Familiarity with Component Data Sheets – Cypress CY6116, CY7C1006, Specifications.

TEXT BOOKS:

1. Digital Design Principles & Practices – John F. Wakerly, PHI/ Pearson Education Asia, 3rd Ed., 2005.
2. A VHDL Primer – J. Bhasker, Pearson Education/ PHI, 3rd Edition.

REFERENCES:

1. Digital System Design Using VHDL – Charles H. Roth Jr., PWS Publications, 2nd edition, 2008.
2. Fundamentals of Digital Logic with VHDL Design – Stephen Borwn and Zvonko Vramesic, McGraw Hill, 2nd Edition., 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04505) LINEAR & DIGITAL IC APPLICATIONS LAB
(Common to ECE, ECM)**

B.Tech III-I Sem. (E.C.E.)	T	P	C
	0	3	2

Minimum Twelve Experiments to be conducted:

Part A (IC Application Lab):

1. OP AMP Applications – Adder, Subtractor, Comparator Circuits.
2. Active Filter Applications – LPF, HPF (first order).
3. Function Generator using OP AMPs.
4. IC 555 Timer – Monostable and Astable Operation Circuit.
5. IC 566 – VCO Applications.
6. Voltage Regulator using IC 723.
7. 4 bit DAC using OP AMP.

Part B (ECAD Lab):

Simulate the internal structure of the following Digital IC's using VHDL / VERILOG and verify the operations of the Digital IC's (Hardware) in the Laboratory

1. Logic Gates- 74XX.
2. Half Adder, Half Subtractor, Full Adder, Full Subtractor & Ripple Carry Adder.
3. 3-8 Decoder -74138 & 8-3 Encoder- 74X148.
4. 8 x 1 Multiplexer -74X151 and 2x4 Demultiplexer-74X155.
5. 4 bit Comparator-74X85.
6. D Flip-Flop 74X74.
7. JK Flip-Flop 74X109.
8. Decade counter-74X90.
9. Universal shift register -74X194.

Equipment required for Laboratories:

1. RPS
2. CRO
3. Function Generator

4. Multi Meters
5. IC Trainer Kits (Optional)
6. Bread Boards
7. Components: - IC741, IC555, IC566, 7805, 7809, 7912 and other essential components.
8. Analog IC Tester

For Software Simulation

1. Computer Systems
2. LAN Connections (Optional)
3. Operating Systems
4. VHDL/ VERILOG
5. FPGAS/CPLDS (Download Tools)

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04506) PULSE & DIGITAL CIRCUITS LAB
(Common to ECE, E Con E, EIE)**

B.Tech III-I Sem. (E.C.E.)	T	P	C
	0	3	2

Minimum Twelve experiments to be conducted:

1. Linear wave shaping.
2. Non Linear wave shaping – Clippers.
3. Non Linear wave shaping – Clampers.
4. Transistor as a switch.
5. Study of Logic Gates & Some applications.
6. Study of Flip-Flops & some applications.
7. Sampling Gates.
8. Astable Multivibrator.
9. Monostable Multivibrator.
10. Bistable Multivibrator.
11. Schmitt Trigger.
12. UJT Relaxation Oscillator.
13. Bootstrap sweep circuit.
14. Constant Current Sweep Generator using BJT.

Equipment required for Laboratories:

- | | | |
|------------------------|---|--------------|
| 1. RPS | - | 0 – 30 V |
| 2. CRO | - | 0 – 20 M Hz. |
| 3. Function Generators | - | 0 – 1 M Hz |
| 4. Components | | |
| 5. Multi Meters | | |

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04601) DIGITAL COMMUNICATIONS**

B.Tech III-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**DIGITIZATION TECHNIQUES FOR ANALOG MESSAGES-I:**

Introduction - Importance of Digitization Techniques for Analog Messages, Pulse Code Modulation (PCM) - Generation and Reconstruction, Quantization Noise, Non-Uniform Quantization and Companding, PCM with Noise, Decoding Noise, Error Threshold, PCM versus Analog Modulation.

UNIT II**DIGITIZATION TECHNIQUES FOR ANALOG MESSAGES-II:**

Delta modulation, Adaptive Delta Modulation, Differential PCM systems (DPCM), Digital Multiplexing-Multiplexers and Hierarchies.

UNIT III**BASE BAND DIGITAL TRANSMISSION-I:**

Digital Signals and Systems – Digital PAM Signals, Transmission Limitations, Power Spectra of Digital PAM, Noise and Errors – Binary Error Probabilities, Matched Filtering, Correlation Detection.

UNIT IV**BASE BAND DIGITAL TRANSMISSION-II:**

Band Limited Digital PAM Systems – Nyquist Pulse Shaping, Optimum Terminal Filters, Correlative Coding.

UNIT V**CHANNEL CODING:**

Error Detection & Correction - Repetition & Parity Check Codes, Interleaving, Code Vectors and Hamming Distance, Forward Error Correction (FEC) Systems, Automatic Retransmission Query (ARQ)

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Systems, Linear Block Codes – Matrix Representation of Block Codes, Convolutional Codes – Convolutional Encoding, Decoding Methods.

UNIT VI**INFORMATION THEORY:**

Information Measure and Encoding - Information Measure, Entropy and Information Rate, Coding for a Discrete Memory Less Channel, Information transmission on a discrete channels - mutual information, Binary Symmetric Channel, Discrete Channel Capacity, Coding for the Binary Symmetric Channels.

UNIT VII**BAND PASS DIGITAL TRANSMISSION-I:**

Digital CW Modulation – Spectral Analysis of Digital Band Pass Signals, Signal Space, Gram-Schmidt Procedure, Coherent Binary Systems – Optimum Binary Detection, Coherent ASK (OOK (on-off keying)), BPSK and FSK, Timing and Synchronization, Interference, Non-Coherent Binary Systems, Non-Coherent FSK, Differentially Coherent PSK.

UNIT VIII**BAND PASS DIGITAL TRANSMISSION-II:**

Quadrature Carrier and M-ary Systems- Quadrature Carrier Systems, M-ary PSK Systems, M-ary QAM Systems, M-ary FSK Systems, and Comparison of Digital Modulation Systems.

TEXT BOOKS:

1. A. Bruce Carlson, & Paul B. Crilly, “Communication Systems – An Introduction to Signals & Noise in Electrical Communication”, McGraw-Hill International Edition, 5th Edition, 2010.
2. Sam Shanmugam, “Digital and Analog Communication Systems”, John Wiley, 2005.

REFERENCES:

1. Bernard Sklar, “Digital Communications”, Prentice-Hall PTR, 2nd edition, 2001.

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2. Simon Haykin, "Communication Systems", Wiley-India edition, 3rd edition, 2010.
3. B.P. Lathi, & Zhi Ding, "Modern Digital & Analog Communication Systems", Oxford University Press, International 4th edition, 2010.
4. Herbert Taub & Donald L Schilling, "Principles of Communication Systems", Tata McGraw-Hill, 3rd Edition, 2009.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04602) MICROPROCESSORS & MICROCONTROLLERS
(Common to CSE, ECE, E Con E, EIE, EEE)**

B.Tech III-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I**INTRODUCTION**

Architecture of 8086 microprocessor, special functions of general purpose registers. 8086 flag register and function of 8086 flags, addressing modes of 8086, instruction set of 8086, assembler directives, simple programs, procedures and macros.

UNIT-II**ASSEMBLY LANGUAGE PROGRAMMING**

Assembly language programs involving logical, branch and call instructions, sorting, evaluation of arithmetic expressions, string manipulation.

UNIT-III**ARCHITECTURE OF 8086 & INTERFACING**

Pin diagram of 8086-Minimum mode and maximum mode of operation, Timing diagram, memory interfacing to 8086 (static RAM and EPROM). Need for DMA. DMA data transfer method. Interfacing with 8237/8257.

UNIT-IV**PROGRAMMABLE INTERFACING DEVICES**

8255 PPI-various modes of operation and interfacing to 8086. interfacing keyboard, displays, 8279 stepper motor and actuators. D/A and A/D converter interfacing, Interrupt structure of 8086, Vector interrupt table. Interrupt service routines. Introduction to DOS and BIOS interrupts. 8259 PIC architecture and interfacing cascading of interrupt controller and its importance.

UNIT-V**SERIAL DATA TRANSFER SCHEMES**

Asynchronous and synchronous data transfer schemes. 8251 USART architecture and interfacing. TTL to RS232C and RS232C to TTL

conversion. Sample program of serial data transfer. Introduction to high-speed serial communications standards, USB.

UNIT-VI

PROGRAMMABLE INTERRUPT CONTROLLERS

PIC 8259, Programming with 8259, Programmable interval timer 8253, Modes of 8253, Programming examples with 8253.

UNIT-VII

8051 MICROCONTROLLER AND ITS PROGRAMMING

Architecture of micro controller-8051 Microcontroller-internal and external memories-counters and timers-synchronous serial-cum asynchronous serial communication-interrupts. Addressing modes of 8051, Instruction set of 8051, Assembly Language Programming examples using 8051.

UNIT-VIII

ADVANCED MICROCONTROLLERS

MCS – 96 Microcontrollers: Important Features, Pin Diagram, Internal Architecture, Memory Map, Addressing Modes, Instruction set. ARM Microcontrollers: ARM Core Architecture, Versions of ARM, Important Features.

TEXT BOOKS:

1. Advanced microprocessor and peripherals-A.K. Ray and K.M.Bhurchandi, 2nd edition, TMH, 2000.
2. Microcontrollers-Deshmukh, Tata Mc-Graw Hill Edition, 2004.
3. Microcontrollers Architecture, programming, interfacing and system Design-Raj kamal, Pearson Education, 2005.

REFERENCES:

1. Microprocessors Interfacing-Douglas V.Hall, 2nd edition, 2007.
2. The 8088 and 8086 Microprocessors- Walter A. Triebel, Avtar Singh, PHI, 4th Edition, 2003.
3. Micro computer system 8066/8088 family Architecture, programming and Design-By Liu and GA Gibson, PHI, 2nd Ed.
4. 8051 Microcontroller-Internals, Instructions, Programming and Interfacing by Subrata Ghoshal, Pearson, 2010.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04603) DIGITAL SIGNAL PROCESSING
(Common to ECE, E Con E, EIE, ECM)**

B.Tech III-II Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT-I**INTRODUCTION**

Introduction to digital signal processing: Discrete time signals and sequences, linear shift invariant systems, stability and causality, linear constant coefficient difference equations. Frequency domain representation of discrete time signals and systems.

UNIT-II**DISCRETE FOURIER SERIES**

Properties of discrete Fourier series, DFS representation of periodic sequences, discrete Fourier transforms: properties of DFT, linear convolution of sequences using DFT, computation of DFT. Relation between Z-Transform and DFS.

UNIT-III**FAST FOURIER TRANSFORMS**

Fast Fourier transforms (FFT)-Radix2 decimation in time and decimation in frequency FFT algorithms, inverse FFT and FFT for composite N.

UNIT-IV**REALIZATION OF DIGITAL FILTERS**

Review of Z-transforms, applications of Z-Transforms, solution of difference equations of digital filters, block diagram representation of linear constant-coefficient difference equations, basic structures of IIR systems, transposed forms, basic structures of FIR systems, system function.

UNIT-V**IIR DIGITAL FILTERS**

Analog filter approximations-Butterworth and chebyshev, design of IIR digital filters from analog filters, design examples: analog-digital transformations, Illustrative Problems.

UNIT-VI**FIR DIGITAL FILTERS**

Characteristics of FIR digital filters, frequency response. Design of FIR digital filters using window techniques, frequency sampling technique, comparison of IIR and FIR filters, Illustrative Problems.

UNIT-VII**MULTIRATE DIGITAL SIGNAL PROCESSING****FUNDAMENTALS:**

Basic sample rate alteration devices, Multirate Structures for sampling rate Converters, Multistage design of decimator and Interpolator, Polyphase Decomposition, Nyquist filters.

UNIT-VIII**APPLICATIONS OF DIGITAL SIGNAL PROCESSING:**

Spectral analysis of nonstationary Signals, Musical Sound processing, signal Compression, Transmultiplexers, Discrete Multitone Transmission of digital data.

TEXT BOOKS:

1. Digital signal processing, principles, Algorithms and applications: John G. Proakis, Dimitris G. Manolakis, Pearson Education/PHI, 4th ed., 2007.
2. Digital signal processing , A computer base approach- Sanjit K Mitra, Tata McGraw Hill, 3rd edition, 2009.
3. Discrete Time Signal Processing-A.V. Oppenheim and R.W. Schaffer, 2nd ed., PHI.

REFERENCES:

1. Digital signal processing: Andreas Antoniou, TATA McGraw Hill, 2006.

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2. A Text book on Digital Signal processing – R S Kaler, M Kulkarni,, Umesh Gupta, I K International Publisthing House Pvt. Ltd.
3. Digital signal processing: M H Hayes, Schaum's outlines, TATA Mc-Graw Hill, 2007.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04604) ELECTRONIC MEASUREMENTS AND
INSTRUMENTATION**

B.Tech III-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I

Performance characteristics of Instruments: Static characteristics, Accuracy, Precision, Resolution, Sensitivity, static and dynamic calibration, Errors in Measurement, and their statistical analysis, dynamic characteristics-speed of Response, fidelity, Lag and dynamic error. DC voltmeters-multirange, range extension/solid state and differential voltmeters, AC voltmeters –multirange, range extension. Thermocouple type RF ammeter, ohm meters, series type, shunt type, multimeter for voltage, current and resistance measurements.

UNIT-II

Signal generator-fixed and variable, AF oscillators, function generators, pulse, random noise, sweep, and arbitrary waveform generators, their standards, specifications and principles of working (Block diagram approach).

UNIT-III

Wave analyzers, Harmonic distortion analyzers, FFT analyzers, and Logic analyzers.

UNIT-IV

Oscilloscopes: Standard specifications of CRO, CRT features, vertical and horizontal amplifiers, horizontal and vertical deflection systems, sweep trigger pulse, delay line, sync selector circuits, probes for CRO – active, passive, and attenuator type, triggered sweep CRO, and Delayed sweep, dual trace/beam CRO, Measurement of amplitude, frequency and phase (Lissajous method).

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UNIT-V

Principles of sampling oscilloscope, storage oscilloscope, and digital storage oscilloscope, Digital frequency counter, time and period measurement, Digital Multimeter (A to D converter used in DMM and its principle).

UNIT-VI

Review of DC Bridges: Wheatstone bridge, Wein Bridge, errors and precautions in using bridges, AC bridges: Measurement of inductance- Maxwell's bridge, Anderson Bridge. Measurement of capacitance- Schearing Bridge. Kelvin Bridge, Q-meter, EMI and EMC, Interference and noise reduction techniques.

UNIT-VII

Sensors and Transducers - Active and passive transducers: Measurement of displacement (Resistance, capacitance, inductance; LVDT) Force (strain gauges) Pressure (piezoelectric transducers) Temperature (resistance thermometers, thermocouples, and thermistors), Velocity, Acceleration, Vibration, pH measurement Signal Conditioning Circuits.

UNIT-VIII

Data Acquisition System, Analogue and digital data recording techniques, strip chart and XY recording methods .Over view of PC Based instrumentation. Bus standards for measuring instruments (GPIB, RS232, USB).

TEXT BOOKS:

1. Electronic instrumentation, second edition - H.S.Kalsi, Tata McGraw Hill, 2004.
2. Modern Electronic Instrumentation and Measurement Techniques – A.D. Helfrick and W.D. Cooper, PHI, 5th Edition, 2002.
3. Electronic Instrumentation & Measurements - David A. Bell, PHI, 2nd Edition, 2003.

REFERENCES:

1. Measurement Systems Application and Design-Ernest O Doebelin and Dhanesh N Manik TMH, 5th Edition, 2009.
2. Electronic Measurement and Instrumentation, Oliver and Cage, TMH.
3. Electronic Test Instruments, Analog and Digital Measurements - Robert A. Witte, Pearson Education, 2nd Ed., 2004.
4. Electronic Measurements & Instrumentations by K. Lal Kishore, Pearson Education - 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

Electronics and Communication Engineering

(9A04605) VLSI DESIGN

(Common to ECE, E Con E, EIE)

B.Tech III-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I

INTRODUCTION

Introduction to IC technology-MOS, PMOS, NMOS, CMOS and BI-CMOS technologies-oxidation, lithiography, diffusion, Ion implantation, metallisation, Encapsulation, probe testing, integrated resistors and capacitors.

UNIT II

BASIC ELECTRICAL PROPERTIES

Basic electrical properties of MOS and BI-CMOS circuits: I_{ds} - V_{ds} relationships, MOS transistor threshold voltage, g_m , g_{ds} , figure of merit; pass transistor, NMOS inverter, various pull-ups, CMOS inverter analysis and design, BI-CMOS inverters.

UNIT III

VLSI CIRCUIT DESIGN PROCESSES

VLSI design flow, MOS layers, stick diagrams, design rules and layout, 2 m CMOS design rules for wires, contacts and transistors layout diagrams for NMOS and CMOS inverters and gates, scaling of MOS circuits, limitations of scaling.

UNIT IV

GATE LEVEL DESIGN

Logic gates and other complex gates, switch logic, alternate gate circuits, basic circuit concepts, sheet resistance R_s and its concept to MOS, area capacitance units, calculations-(Micro)-delays, driving large capacitive loads, wiring capacitances, fan-in and fan-out, choice of layers.

UNIT V

SUB SYSTEM DESIGN

Sub system design, shifters, adders, ALUs, multipliers, parity generators, comparators, zero/one detectors, counters, high density memory elements.

UNIT VI**SEMICONDUCTOR INTEGRATED CIRCUIT DESIGN**

PLAs, FPGAs, CPLDs, standard cells, programmable array logic, design approach.

UNIT VII**VHDL SYNTHESIS**

VHDL synthesis, circuit design flow, circuit synthesis, simulation, layout, design capture tools, design verification tools, test principles.

UNIT VIII**CMOS TESTING**

CMOS testing need for testing, test principles, design strategies for test, chip level test techniques, system-level test techniques, layout design for improved testability.

TEXT BOOKS:

1. Essentials of VLSI circuits and systems-kamran Eshraghian, Eshraghian Douglas and A. pucknell, PHI, 2005 Edition.
2. Principles of CMOS VLSI design-Weste and EShraghian, Pearson Education, 1999.

REFERENCES:

1. Introduction to VLSI circuits and systems-John P.Uyemura, John Wiley, 2003.
2. Digital Integrated circuits-John M. Rabaey, PHI, EEE, 1997.
3. Modern VLSI design-Wayne wolf, Pearson Education, 3rd Edition, 1997.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04606) MICROWAVE ENGINEERING**

B.Tech III-II Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT-I**MICROWAVE TRANSMISSION LINES - I:**

Introduction, Microwave spectrum and bands, applications of Microwaves. Rectangular Waveguides-Solution of Wave Equation in Rectangular Coordinates, TE/TM mode analysis, Expressions for fields, Characteristic equation and cutoff frequencies, filter characteristics, dominant and degenerate modes, sketches of TE and TM mode fields in the cross-section. Mode characteristics - Phase and Group velocities, wavelengths and impedance relations, Illustrative Problems.

UNIT-II**MICROWAVE TRANSMISSION LINES - II:**

Rectangular Waveguides – Power Transmission and Power Losses, Impossibility of TEM Modes, Micro strip lines-introduction, Z_0 relations, effective dielectric constant, losses, Q-factor, Cavity resonators-introduction, Rectangular and cylindrical cavities, dominant modes and resonant frequencies, Q-factor and coupling coefficients, Illustrative Problems.

UNIT-III**WAVEGUIDE COMPONENTS AND APPLICATIONS- I:**

Coupling mechanisms- probe, loop, aperture types. Wave guide discontinuities - waveguide Windows, tuning screws and posts, matched loads. Waveguide attenuators - resistive card, rotary vane Attenuators; waveguide phase shifters-dielectric, rotary vane phase shifters. Wave guide multiport junctions-E plane and H plane Tees, Magic Tee, Directional couplers-2 hole, Bothe hole types, Illustrative Problems.

2009-10**UNIT-IV****WAVE GUIDE COMPONENTS AND APPLICATIONS-II:**

Ferrites-composition and characteristics, Faraday rotation; Ferrite components-Gyrator, Isolator, Circulator.Scattering Matrix-Significance, Formulation and properties. S Matrix calculations for 2-port junction, E plane and H plane Tees, Magic Tee, Directional coupler, circulator and Isolator, Illustrative Problems.

UNIT-V**MICROWAVE TUBES-I:**

Limitations and losses of conventional tubes at microwave frequencies. Microwave tubes-O type and M type classifications. O type tubes: 2 cavity klystrons-structure, Reentrant cavities, velocity modulation process and Applegate diagram, bunching process and small signal theory-Expressions for O/P power and efficiency. Reflex Klystrons-structure, Velocity Modulation, Applegate diagram, mathematical theory of bunching, power output, efficiency, oscillating modes and O/P characteristics, Effect of Repeller Voltage on Power O/P, Illustrative Problems.

UNIT-VI**HELIX TWTS:**

Significance, types and characteristics of slow wave structures; structure of TWT and amplification process (qualitative treatment), suppression of oscillations, gain considerations.

M -TYPE TUBES: Introduction, cross field effects, Magnetrons-different types, cylindrical travelling wave magnetron-Hull cutoff and Hartree conditions, modes of resonance and PI-mode operation, separation of PI-mode, O/P characteristics, Illustrative Problems.

UNIT-VII**MICROWAVE SOLID STATE DEVICES:**

Introduction, classification, applications, Transfer Electronic Devices, Gunn diode - principles, RWH theory, characteristics, basic modes of operation - Gunn oscillation modes. LSA Mode, Varactor Diode, Parametric Amplifier, Introduction to Avalanche Transit time devices (brief treatment only).

2009-10

UNIT-VIII**MICROWAVE MEASUREMENTS:**

Description of Microwave bench-different blocks and their features, errors and precautions; Microwave power measurement-Bolometers, Measurement of attenuation, frequency standing wave measurements – measurement of low and high VSWR, cavity-Q, impedance measurements.

TEXT BOOKS:

1. Microwave devices and circuits-Samuel Y. Liao, Pearson, 3rd Edition, 2003.
2. Microwave principles-Herbert J.Reich, J.G.Skalnik, P.F.Ordung and H.L.Krauss, CBS publishers and distributors, New Delhi, 2004.

REFERENCES:

1. Foundations for microwave engineering-R.E.Collin, IEEE press, John Wiley, 2nd edition, 2002.
2. Microwave circuits and passive devices-M.L.Sisodia and G.S.Raghuvanshi,Wiley Eastern Ltd., New age International publishers Ltd., 1995.
3. Microwave engineering passive circuits-Peter A.Rizzi, PHI, 1999.
4. Electronic and Radio Engineering-F.E.Terman, McGraw-Hill, 4th Edition, 1995.
5. Microwave Engineering – A. Das, TMH, 2nd ed., 2009.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04607) ANALOG & DIGITAL COMMUNICATIONS LAB**

B.Tech III-II Sem. (E.C.E.)	T	P	C
	0	3	2

Minimum Twelve Experiments to be conducted: (Six from each part A & B)

Part A (Analog Communication Lab):

1. Amplitude modulation and demodulation.
2. Frequency modulation and demodulation.
3. Characteristics of Mixer.
4. Pre-emphasis & de-emphasis.
5. Pulse Amplitude Modulation and demodulation.
6. Pulse Width Modulation and demodulation.
7. Pulse Position Modulation and demodulation.
8. Radio Receiver measurements – Sensitivity, Selectivity, & Fidelity.

Part B (Digital Communication Lab):

1. Sampling Theorem – verification.
2. Time division multiplexing.
3. Pulse Code Modulation.
4. Delta modulation.
5. Frequency shift keying - Modulation and Demodulation.
6. Phase shift keying - Modulation and Demodulation.
7. Differential phase shift keying - Modulation and Demodulation.
8. QPSK - Modulation and Demodulation.

Equipment required for Laboratories:

1. RPS - 0 – 30 V
2. CRO - 0 – 20 M Hz.
3. Function Generators - 0 – 1 M Hz
4. RF Generators - 0 – 1000 M Hz. / 0 – 100 M Hz.
5. Multimeters
6. Lab Experimental kits for Communication
7. Components
8. Radio Receiver/TV Receiver Demo kits or Trainees

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9AHS601) ADVANCED ENGLISH LANGUAGE
COMMUNICATION SKILLS LAB
(Common to ECE, E Con E, ECM, EIE, EEE, ME, AE)**

B.Tech III-II Sem. (E.C.E.)

T	P	C
0	3	2

1. Introduction

The Advanced English Language Skills Lab introduced at the 3rd year B.Tech level is considered essential for the student for focusing on his/her career. At this stage it is imperative for the student to start preparing for the ever growing competition in the job market. In this scenario, in order to be on par with the best, he/she needs to improve his/her Communication and soft skills

This course focuses on the practical aspects of English incorporating all the four (LRSW) skills relevant to the requirements of the prospective employers in view of globalization. The proposed course will enable the students to perform the following:

- Intensive reading to improve comprehension and communication
- Attentive listening for better understanding
- Write project/research/technical reports
- Write Resume' to attract attention
- Discuss ideas / opinions for better solutions
- Face interviews confidently
- Gather information, organize ideas, and present them effectively before an audience
- To help the students cultivate the habit of reading passages from the computer monitor, thus providing them with the required ability to face computer-based competitive exams such GRE, TOEFL, CAT, GMAT etc.

2. Objectives:

Keeping in mind the previous exposure of the student to English, this lab focuses on improving the student's proficiency in English at all levels. The lab intends to train students to use language effectively, to participate in group discussions, to help them face interviews, and sharpen public speaking skills and enhance the confidence of the student by exposing him/her to various situations and contexts which he/she would face in his/her career

3 Syllabus

The following course content is prescribed for the Advanced Communication Skills Lab:

Reading Comprehension -- Reading for facts, guessing meanings from context, speed reading, scanning, skimming for building vocabulary(synonyms and antonyms, one word substitutes, prefixes and suffixes, idioms and phrases.)

Listening Comprehension-- Listening for understanding, so as to respond relevantly and appropriately to people of different backgrounds and dialects in various personal and professional situations.

Technical Report Writing—Types of formats and styles, subject matter, organization, clarity, coherence and style, data-collection, tools, analysis

Resume' Writing—Structure, format and style, planning, defining the career objective, projecting one's strengths, and skills, creative self marketing, cover letter

Group Discussion-- Communicating views and opinions, discussing, intervening. providing solutions on any given topic across a cross-section of individuals,(keeping an eye on modulation of voice, clarity, body language, relevance, fluency and coherence) in personal and professional lives.

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Interview Skills—Concept and process, pre-interview planning, mannerisms, body language, organizing, answering strategies, interview through tele and video-conferencing

Technical Presentations (Oral)— Collection of data, planning, preparation, type, style and format ,use of props, attracting audience, voice modulation, clarity, body language, asking queries.

4. Minimum Requirements

The English Language Lab shall have two parts:

The Computer aided Language Lab for 60 students with 60 systems, one master console, LAN facility and English language software for self-study by learners.

The Communication Skills Lab with movable chairs and audio-visual aids with a P.A System, a TV, A digital stereo-audio and video system, Camcorder etc

System Requirement (Hardware Component):

Computer network with LAN with a minimum of 60 multimedia systems with the following specifications:

P-IV Processor, Speed-2.8 GHz, RAM_512 MB minimum, Hard Disk-80 GB, Headphones

Prescribed Software: GLOBARENA

Books Suggested for English Language Lab Library (to be located within the lab in addition to the CDs of the text book which are loaded on the systems):

1. **Technical writing and professional communication, Huckin and Olsen** Tata Mc Graw-Hil 2009.
2. **Speaking about Science, A Manual for Creating Clear Presentations by Scott Morgan and Barrett Whitener, Cambridge University press, 2006**
3. **Books on TOEFL/GRE/GMAT/CAT/ IELTS by Barron's/DELTA/Cambridge University Press.**
4. **Handbook for Technical Writing** by David A McMurrey & Joanne Buckely CENGAGE Learning 2008

5. **Technical Communication** by Meenakshi Raman & Sangeeta Sharma, Oxford University Press 2009.
6. **The ACE of Soft Skills** by Gopal Ramesh and Mahadevan Ramesh, Pearson Education, 2010
7. **Cambridge English for Job-Hunting** by Colm Downes, Cambridge University Press, 2008
8. **Resume's and Interviews** by M.Ashraf Rizvi, Tata Mc Graw-Hill, 2008
9. **From Campus To Corporate** by KK Ramachandran and KK Karthick, Macmillan Publishers India Ltd, 2010
10. **English Language Communication : A Reader cum Lab Manual** Dr A Ramakrishna Rao, Dr G Natanam & Prof SA Sankaranarayanan, Anuradha Publications, Chennai 2008.
11. **Managing Soft Skills** by K R Lakshminarayan and T.Muruguvel, Sci-Tech Publications, 2010
12. **Business Communication** by John X Wang, CRC Press, Special Indian Edition, 2008

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9AHS701) MANAGEMENT SCIENCE
(Common to ECE, E Con E, EIE)**

B.Tech IV-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I**INTRODUCTION TO MANAGEMENT:**

Concepts of Management and organization- nature, importance and Functions of Management, Taylor's Scientific Management Theory, Fayol's Principles of Management, Mayo's Hawthorne Experiments, Maslow's Theory of Human Needs, Douglas McGregor's Theory X and Theory Y, Herzberg's Two-Factor Theory of Motivation, Systems Approach to Management, Leadership Styles, Social responsibilities of Management.

UNIT II**DESIGNING ORGANIZATIONAL STRUCTURES:**

Basic concepts related to Organisation - Departmentation and Decentralisation, Types of mechanistic and organic structures of organisation (Line organization, Line and staff organization, functional organization, Committee organization, matrix organization, Virtual Organisation, Cellular Organisation, team structure, boundaryless organization, inverted pyramid structure, lean and flat organization structure) and their merits, demerits and suitability.

UNIT III**OPERATIONS MANAGEMENT:**

Principles and Types of Plant Layout-Methods of production (Job, batch and Mass Production), Work Study -Basic procedure involved in Method Study and Work Measurement- Statistical Quality Control: chart, R chart, $\bar{c}$ chart, p chart, (simple Problems), Acceptance Sampling, Deming's contribution to quality.

2009-10**UNIT IV****MATERIALS MANAGEMENT:**

Objectives, Need for Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Stores Records.

Marketing: Functions of Marketing, Marketing Mix, Marketing Strategies based on Product Life Cycle, Channels of distribution

UNIT V**HUMAN RESOURCES MANAGEMENT (HRM):**

Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs.PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating.

UNIT VI**PROJECT MANAGEMENT (PERT/CPM):**

Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing. (simple problems)

UNIT VII**STRATEGIC MANAGEMENT:**

Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives.

UNIT VIII**CONTEMPORARY MANAGEMENT PRACTICES:**

Basic concepts of MIS, End User Computing, Materials Requirement Planning (MRP), Just-In-Time (JIT) System, Total Quality Management (TQM), Six sigma and Capability Maturity Model (CMM) Levels, Supply Chain Management, Enterprise Resource Planning (ERP), Performance Management, Business Process outsourcing

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(BPO), Business Process Re-engineering and Bench Marking, Balanced Score Card.

TEXT BOOKS:

1. Aryasri: Management Science, TMH, 2004.
2. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.

REFERENCES:

1. Kotler Philip & Keller Kevin Lane: Marketing Mangement 12/e, PHI, 2005.
2. Koontz & Weihrich: Essentials of Management, 6/e, TMH, 2005.
3. Thomas N.Duening & John M.Ivancevich Management—Principles and Guidelines, Biztantra, 2003.
4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2004.
5. Memoria & S.V.Gauker, Personnel Management, Himalaya, 25/e, 2005
6. Samuel C.Certo: Modern Management, 9/e, PHI, 2005
7. Schermerhorn, Capling, Poole & Wiesner: Management, Wiley, 2002.
8. Parnell: Strategic Management, Biztantra, 2003.
9. Lawrence R Jauch, R.Gupta & William F.Glueck: Business Policy and Strategic Management, Frank Bros., 2005.
10. L.S.Srinath: PERT/CPM, Affiliated East-West Press, 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04701) EMBEDDED REALTIME OPERATING SYSTEMS
(Common to ECE, CSSE, E Con E, EIE)**

B.Tech IV-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I**INTRODUCTION**

History of Embedded Systems, Major Application Areas of Embedded Systems, Purpose of Embedded Systems, Core of the Embedded System, Sensors and Actuators, Communication Interface, Embedded Firmware.

UNIT II**HARDWARE SOFTWARE Co-DESIGN and PROGRAMME MODELLING**

Characteristics of an Embedded System, Quality Attributes of Embedded Systems, Fundamental Issues in Hardware Software Co-Design, Computational Models in Embedded Design, Introduction to Unified Modeling Language (UML), Hardware Software Trade-offs.

UNIT III**EMBEDDED HARDWARE DESIGN AND DEVELOPMENT**

Analog Electronic Components, Digital Electronic Components, VLSI and Integrated Circuit Design, Electronic Design Automation (EDA) Tools, Embedded Firmware Design Approaches, Embedded Firmware Development Languages.

UNIT IV**REAL-TIME OPERATING SYSTEMS (RTOS) BASED EMBEDDED SYSTEM DESIGN**

Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling, Threads, Processes and Scheduling :Putting them Altogether, Task Communication, Task Synchronization, Device Drivers, How to Choose an RTOS.

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UNIT V**DEVICES AND COMMUNICATION BUSES FOR DEVICES NETWORK**

IO Types and Examples, Serial Communication Devices, Parallel Device Ports, Sophisticated Interfacing Features in Device Ports, Wireless Devices, Timer and Counting Devices, Watchdog Timer, Real Time Clock, Networked Embedded Systems, Serial Bus Communication Protocols, Parallel Bus Device Protocols- Parallel Communication Network Using ISA, PCI, PCI-X and Advanced Buses, Internet Enabled Systems- Network Protocols, Wireless and Mobile System Protocols.

UNIT VI**PROGRAM MODELING CONCEPTS**

Program Models, DFG Models, state Machine Programming Models for Event-controlled Program Flow, Modeling of Multiprocessor Systems, UML Modeling.

UNIT VII**REAL TIME OPERATING SYSTEMS**

OS Services, Process Management, Timer .Functions, Event Functions, Memory Management, Device, File and IO Subsystems Management, Interrupt Routines in RTOS Environment and Handling of Interrupt Source Calls, Real-time Operating Systems, Basic-Design an RTOS, RTOS Task Scheduling Models, Interrupt Latency and Response of the Tasks as Performance Matrices, OS Security Issues.

UNIT VIII**DESIGN EXAMPLES AND CASE STUDIES OF PROGRAM MODELING AND PROGRAMMING WITH RTOS-2**

Case study of Communication between Orchestra Robots, Embedded Systems in Automobile, Case study of an Embedded System for an Adaptive Cruise Control(ACC) System in a Car, Case study of an Embedded System for a Smart Card, Case study of a Mobile Phone Software for Key Inputs.

TEXT BOOKS:

1. Introduction to Embedded System- Shibu KV, Mc-Graw Hill Higher Edition.
2. Embedded Systems Architecture, Programming and Design- Raj Kamal, Second Edition, McGraw-Hill Companies.
3. Embedded System Design by Peter Marwedel, Springer.

REFERENCES:

1. Embedded System Design – A Unified Hardware/Software Introduction-Frank Vahid, Tony D. Givargis, John Wiley, 2002.
2. Embedded/ Real Time Systems-KVKK Prasad, Dreamtech Press, 2005.
3. An Embedded Software Primer- David E. Simon, Pearson Ed. 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A05506) COMPUTER NETWORKS
(Common to ECE, EIE)**

B.Tech IV-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I

Introduction: Network Hardware, Network Software, References Models. The Physical Layer: The Theoretical Basis for Data Communication Guided Transmission Media, Communication Satellites, The public Switched Telephone Network- The Local Loop: Modern ADSL, and wireless, Trunks and Multiplexing, Switching

UNIT II

The Data Link Layer: Data link Layer Design Issues, Elementary Data Link Protocols, Sliding Window Protocols.

UNIT III

The Medium Access Control Sublayer: The Channel allocation Problem, Multiple Access protocols, Ethernet- Ethernet Cabling, Manchester Encoding, The Ethernet MAC Sublayer Protocol. The Binary Exponential Backoff Algorithm, Ethernet Performance, Switched Ethernet, Fast Ethernet. Wireless Lans- The 802.11 Protocol Stack, the 802.11 Physical Layer, the 802.11 MAC SubLayer Protocol, The 802.11 Frame Structure.

UNIT IV

The Network Layer: Network Layer Design Issues, Routing Algorithms, Congestion Control Algorithms.

UNIT V

Internetworking, The Network Layer in the Internet.

UNIT VI

The Transport Layer: The Transport Service, Elements of Transport Protocols, The Internet Transport Protocols: UDP, The Internet Transport Protocols: TCP.

UNIT VII

The Application Layer: DNS-The Domain Name System, Electronic Mail. The World Wide Web, Multimedia.

UNIT VIII

Network Security: Cryptography, Symmetric-Key Algorithms, Public-Key Algorithms, Digital Signatures.

TEXT BOOKS:

1. Computer Networks, Andrew S. Tanenbaum, 4e, Pearson Education.

REFERENCES:

1. Computer Communications and Networking Technologies, Michael A. Gallo, William M. Hancock, Cengage Learning.
2. Computer Networks-Principles, Technologies and Protocols for Network Design, Natalia Olifer, Victor Olifer, Wiley India.
3. Data Communications and Networking, Behrouz A. Forouzan, 4th Edition, Tata McGraw Hill.
4. Understanding Communications and Networks, 3rd Edition, W.A.Shay, Cengage Learning.
5. Computer and Communication Networks, Nader F. Mir, Pearson Education.
6. Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose, K.W.Ross, 3rd Edition, Pearson Education.
7. Data and Computer Communications, G.S.Hura and M.Singhal, CRC Press, Taylor and Francis Group.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04702) OPTICAL COMMUNICATIONS**

B.Tech IV-I Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT-I

Introduction and Optical fiber waveguides: Historical Development, The General System, Advantages of Optical Fiber Communications, Ray Theory transmission, Electromagnetic mode theory for Optical Propagation, Cylindrical Fiber.

UNIT-II

Single mode fibers, Fiber Materials - Fiber Fabrication, Mechanical Properties of Fibers, Fiber Optic Cables.

UNIT-III

Attenuation, Material Absorption Losses in Silica Glass Fibers, Linear Scattering Losses, Fiber Bend Loss, Dispersion, Chromatic dispersion, Intermodal dispersion, Overall fiber dispersion, Polarization.

UNIT-IV

Fiber alignment and joint loss, Fiber Splices, Fiber Connectors, Fiber Couplers, Optical Isolators and Circulators.

UNIT-V

Light Emitting Diodes (LEDs): LED Structures, Light Source Materials, Quantum efficiency and LED Power, Modulation of LED. LASER Diodes- Laser Diode Modes and Threshold Conditions, Laser Diode Rate Equations, External Quantum Efficiencies, Resonant Frequencies.

UNIT-VI

Power launching and Coupling - Source to Fiber Power Launching, Lensing schemes for Coupling Improvement, fiber-to-fiber Joints, LED coupling to single mode fibers, Fiber Splicing, Optical fiber connectors. Photo Detectors – Physical principles of photo diodes,

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photo detector noise, detector response time, avalanche multiplication noise, structures for InGaAs APDs, temperature effect on avalanche gain, comparisons of photo detectors.

UNIT-VII

Digital Links: Point to point links, power penalties, error control.

Analog links: Over-view of analog links, carrier to noise ratio, multichannel transmission techniques, RF over fiber, radio over fiber links.

UNIT-VIII

WDM Concepts and components: Over-view, Passive optical couplers, Isolators & circulators, Fiber grating filters, dielectric thin film filters, Phased array based devices, Diffraction gratings, Active optical components, tunable light sources.

TEXT BOOKS:

1. Optical fiber communications- Gerd keiser, McGraw Hill International Edition, 4th Edition, 2010.
2. Optical fiber communications-John M. Senior, PHI, 3rd Edition, 2010.

REFERENCES:

1. Principles and Applications of Optical Communications, Max Ming-Kang Liu, TMH, 2010.
2. Text book on optical fiber communication and its applications- S.C.Gupta, PHI, 2005.
3. Fundamentals of Optical Fiber communications, Satish Kumar, PHI, 2009.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04703) RADAR SYSTEMS
(ELECTIVE – I)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**BASICS OF RADAR:**

Introduction, Maximum Unambiguous Range, Simple form of Radar Equation, Radar Block Diagram and Operation, Radar Frequencies and Applications, Prediction of Range Performance, Minimum Detectable Signal, Receiver Noise, Modified Radar Range Equation, Illustrative Problems.

UNIT II**RADAR EQUATION:**

SNR, Envelope Detector, False Alarm Time and Probability, Integration of Radar Pulses, Radar Cross Section of Targets (simple targets - sphere, cone-sphere), Transmitter Power, PRF and Range Ambiguities, System Losses (qualitative treatment), Illustrative Problems.

UNIT III**CW AND FREQUENCY MODULATED RADAR:**

Doppler Effect, CW Radar – Block Diagram, Isolation between Transmitter and Receiver, Non-zero IF Receiver, Receiver Bandwidth Requirements, Applications of CW radar, Illustrative Problems.

UNIT IV

FM-CW Radar, Range and Doppler Measurement, Block Diagram and Characteristics (Approaching/ Receding Targets), FM-CW altimeter, Multiple Frequency CW Radar.

UNIT V**MTI AND PULSE DOPPLER RADAR:**

Introduction, Principle, MTI Radar with - Power Amplifier Transmitter and Power Oscillator Transmitter, Delay Line Cancellers – Filter Characteristics, Blind Speeds, Double Cancellation, And Staggered PRFs. Range Gated Doppler Filters, MTI Radar Parameters, Limitations to MTI Performance, MTI versus Pulse Doppler radar.

UNIT VI**TRACKING RADAR:**

Tracking with Radar, Sequential Lobing, Conical Scan, Monopulse Tracking Radar – Amplitude Comparison Monopulse (one- and two-coordinates), Phase Comparison Monopulse, Tracking in Range, Acquisition and Scanning Patterns, Comparison of Trackers.

UNIT VII**DETECTION OF RADAR SIGNALS IN NOISE:**

Introduction, Matched Filter Receiver – Response Characteristics and Derivation, Correlation Function and Cross-correlation Receiver, Efficiency of Non-matched Filters, Matched Filter with Non-white Noise.

UNIT VIII**RADAR RECEIVERS:**

Noise Figure and Noise Temperature, Displays – types. Duplexers – Branch type and Balanced type, Circulators as Duplexers. Introduction to Phased Array Antennas – Basic Concepts, Radiation Pattern, Beam Steering and Beam Width changes, Series versus Parallel Feeds, Applications, Advantages and Limitations.

TEXT BOOKS:

1. Introduction to Radar Systems – Merrill I. Skolnik, TMH Special Indian Edition, 2nd Edition, 2007.

REFERENCES:

1. Introduction to Radar Systems – Merrill I. Skolnik, 3rd Edition, Tata McGraw-Hill, 2001.
2. Radar Principals, Technology, Applications – Byron Edde, Pearson Education, 2004.
3. Radar Principles – Peebles, Jr., P.Z. Wiley, NweYork, 1998.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04704) DSP PROCESSORS & ARCHITECTURES
(Common to ECE, ECM)
(ELECTIVE – I)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**INTRODUCTION TO PROGRAMMABLE DSPs:**

Multiplier & Multiplier accumulator, Modified bus structures & memory access schemes in P – DSPs, Multiple access memory, Multiported memory, VLIW architecture, Pipelining, Special addressing modes in P–DSPs, On chip peripherals.

UNIT II**COMPUTATIONAL ACCURACY IN DSP****IMPLEMENTATIONS:**

Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.

UNIT III**ARCHITECTURES FOR PROGRAMMABLE DSP DEVICES:**

Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation Unit, Programmability and Program Execution, Speed Issues, Features for External interfacing.

UNIT IV**PROGRAMMABLE DIGITAL SIGNAL PROCESSORS:**

Commercial Digital signal-processing Devices, Data Addressing modes of TMS320C54XX DSPs, Data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX Processors, Program Control, TMS320C54XX instructions and Programming, On-Chip

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Peripherals, Interrupts of TMS320C54XX processors, Pipeline Operation of TMS320C54XX Processors.

UNIT V

IMPLEMENTATIONS OF BASIC DSP ALGORITHMS:

The Q-notation, FIR Filters, IIR Filters, Interpolation Filters, Decimation Filters, PID Controller, Adaptive Filters, 2-D Signal Processing.

UNIT VI

IMPLEMENTATION OF FFT ALGORITHMS:

An FFT Algorithm for DFT Computation, A Butterfly Computation, Overflow and scaling, Bit-Reversed index generation, An 8-Point FFT implementation on the TMS320C54XX, Computation of the signal spectrum.

UNIT VII

INTERFACING MEMORY AND I/O PERIPHERALS TO PROGRAMMABLE DSP DEVICES:

Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA). A Multichannel buffered serial port (McBSP), McBSP Programming, a CODEC interface circuit, CODEC programming, A CODEC-DSP interface example.

UNIT VIII

RECENT TRENDS IN DSP SYSTEM DESIGN:

An over-view of the application nodes on DSP systems, An over-view of open multimedia applications platform (OMAP), An Introduction to FPGA, Design flow for an FPGA based system design, Cad tools for FPGA based system design, soft core processors, FPGA based DSP system design, New algorithms for Implementation of filters in VLSI, Distributed arithmetic algorithm, Case studies, Comparison of the performances of the systems designed using FPGAs and digital signals processors.

2009-10

TEXT BOOKS:

1. Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2. Digital Signal Processors, Architecture, Programming and Applications – B. Venkata Ramani and M. Bhaskar, TMH, 2004.

REFERENCES:

1. Digital Signal Processing – Jonathan Stein, John Wiley, 2005.
2. DSP Processor Fundamentals, Architectures & Features – Lapsley et al. S. Chand & Co, 2000.

2009-10

**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A05505) OPERATING SYSTEMS
(Common to ECE, E Con E, EIE)
(ELECTIVE – I)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**OPERATING SYSTEMS OVERVIEW:**

Operating systems functions, Overview of computer operating systems, protection and security, distributed systems, special purpose systems, operating systems structures: operating system services and systems calls, system programs, operating system structure, operating systems generation.

UNIT II**PROCESS MANAGEMENT:**

Process concepts, threads, scheduling-criteria, algorithms, their evaluation, thread scheduling, case studies UNIX, Linux, Windows.

UNIT III**CONCURRENCY:**

Process synchronization, the critical-section problem, Peterson's Solution, synchronization Hardware, semaphores, classic problems of synchronization, monitors, Synchronization examples, atomic transactions. Case studies UNIX, Linux, Windows.

UNIT IV**MEMORY MANAGEMENT:**

Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement, algorithms, Allocation of frames, Thrashing case studies UNIX, Linux, Windows.

2009-10

UNIT V**PRINCIPLES OF DEADLOCK:**

System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock.

UNIT VI**FILE SYSTEM INTERFACE:**

The concept of a file, Access Methods, Directory structure, File system mounting, file sharing, protection. File System implementation: File system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance, case studies. UNIX, Linux, Windows

UNIT VII**MASS-STORAGE STRUCTURE:**

Overview of Mass-storage structure, Disk structure, disk attachment, disk scheduling, swap-space management, RAID structure, stable-storage implementation, Tertiary storage structure. I/O systems: Hardware, application I/O interface, kernel I/O subsystem, Transforming I/O requests to Hardware operations, STREAMS, performance.

UNIT VIII**PROTECTION:**

Protection, Goals of Protection, Principles of Protection, Domain of protection Access Matrix, Implementation of Access Matrix, Access control, Revocation of Access Rights, Capability- Based systems, Language – Based Protection, Security: The Security problem, program threats, system and network threats cryptography as a security tool, user authentication, implementing security defenses, firewalling to protect systems and networks, computer –security classifications, case studies UNIX, Linux, Windows.

TEXT BOOKS:

1. Operating System Concepts- Abraham Silberchatz, Peter B. Galvin, Greg Gagne, 8th edition, John Wiley.

2. Operating systems- A Concept based Approach-D.M.Dhamdhare, 2nd Edition, TMH.

REFERENCES:

1. Operating Systems – Internals and Design Principles, Stallings, 6th edition–2009, Pearson education.
2. Modern Operating Systems, Andrew S Tanenbaum 2nd edition, PHI.
3. Operating Systems, S.Haldar, A.A.Aravind, Pearson education.
4. Operating Systems, A.S.Godbole, 2nd Edition, TMH.
5. An Introduction to Operating Systems, P.C.P. Bhatt, PHI.
6. Operating Systems, G.Nutt, N.Chaki and S.Neogy, 3rd Edition, Pearson Education.
7. Operating Systems, R.Elmasri, A,G.Carrick and D.Levine, McGraw Hill.

2009-10

**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A10802) BIO-MEDICAL INSTRUMENTATION
(ELECTIVE – II)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I

Components of Medical Instrumentation System, Bio – amplifier, Static and dynamic characteristics of medical instruments, Biosignals and characteristics, Problems encountered with measurements from human beings.

UNIT II

Organisation of cell, Derivation of Nernst equation for membrane Resting Potential Generation and Propagation of Action Potential, Conduction through nerve to neuro-muscular junction.

UNIT III

Bio Electrodes – Biopotential Electrodes-External electrodes, Internal Electrodes, Biochemical Electrodes.

UNIT IV

Mechanical function, Electrical Conduction system of the heart, Cardiac cycle, Relation between electrical and mechanical activities of the heart.

UNIT V

Cardiac Instrumentation Blood pressure and Blood flow measurement, Specification of ECG machine, Einthoven triangle, Standard 12-lead configurations, Interpretation of ECG waveform with respect to electro mechanical activity of the heart, Therapeutic equipment, Pacemaker, Defibrillator, Shortwave diathermy, Hemodialysis machine.

UNIT VI

Neuro-Muscular Instrumentation Specification of EEG and EMG machines, Electrode placement for EEG and EMG recording, Interpretation of EEG and EMG.

UNIT VII

Respiratory Instrumentation Mechanism of respiration, Spirometry, Pneumotachograph Ventilators.

UNIT VIII

Patient electrical safety, types of hazards, natural protective mechanism, leakage current, patient isolation, hazards in operation rooms, grounding conditions in hospital environment.

TEXT BOOKS:

1. Biomedical Instrumentation and Measurements – Leslie Cromwell and F.J. Weibell, E.A. Pfeiffer, PHI, 2nd Ed, 1980.
2. Medical Instrumentation, Application and Design – John G. Webster, John Wiley, 3rd Ed., 1998.

REFERENCES:

1. Principles of Applied Biomedical Instrumentation – L.A. Geoddes and L.E. Baker, John Wiley, 1975.
2. Hand-book of Biomedical Instrumentation – R.S. Khandpur, TMH, 2nd Ed., 2003.
3. Biomedical Telemetry – Mackay, Stuart R., John Wiley, 2nd edition, 1968.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04705) T.V. ENGINEERING
(ELECTIVE – II)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I**INTRODUCTION:**

TV transmitter and receivers, synchronization, Television pictures: Geometric form and aspect Ratio, image continuity, interlaced scanning, picture Resolution, composite video signal: Horizontal and vertical sync, Scanning sequence, color signal generation and encoding: perception of brightness and colors, additive color mixing, video signals for colors, Luminance signal, color difference signals, encoding of color difference signals, formation of Chrominance signals, PAL encoder.

UNIT-II**TV SIGNAL TRANSMISSION AND PROPAGATION:**

Picture signal transmission, positive and negative modulation, VSB transmission, sound signal transmission, standard channel BW, TV transmitter, TV signal propagation, Interference, TV broadcast channels, TV transmission antennas.

UNIT-III**TV CAMERAS:**

Camera tube types, Vidicon, silicon diode array vidicon, monochrome TV camera, color camera, CCD image sensors.

UNIT-IV**PICTURE TUBES:**

Monochromatic picture tube, electrostatic focusing, beam deflection, picture tube characteristics and specifications, color picture tubes, TV standards: American 525 line B&W TV system, NTSC color system, 625-line monochrome system, bPAL color system, TV standards.

UNIT-V**MONOCHROME TV RECEIVER:**

RF tuner, IF sub system, video amplifier, sound section, sync separation and processing, deflection circuits, scanning circuits, PAL-D color receiver: electron tuners, IF sub system, Y-signal channel, Chroma decoder, separation of U & V color phasors, synchronous demodulators, sub carrier generation, Raster circuits.

UNIT-VI**VISION IF SUB SYSTEM:**

AGC, Noise cancellation, video and inter carrier sound signal detection, vision IF sub system of Black & White receivers, color Receivers IF sub system. Receiver sound system: FM detection, FM sound detectors, typical applications. TV receiver tuners: tuner operation, VHF and UHF tuners, digital tuning techniques, Remote control of receiver functions.

UNIT-VII**COLOR SIGNAL DECODING:**

PAL-D decoder, chroma signal amplifiers, separation of U & V signals, color Burst separation, Burst phase discriminator, ACC amplifier, Reference Oscillator, Indent and color killer circuits, RO phase shift and 180° PAL-switch circuitry, U & V demodulators, color signal mixing.

UNIT-VIII**SYNC.SEPARATION, AFC AND DEFLECTION****OSCILLATORS:**

Synchronous separation, K Noise in sync. Pulses, separation of frame and line sync. Pulses, AFC, single ended AFC circuit. Deflection oscillators, deflection drive IC's, Receiver antennas, Digital TV digital satellite TV, direct to home satellite TV, digital TV receiver, digital terrestrial TV.

TEXT BOOKS:

1. Modern television practice-Principles, Technology and service-R.R. Gulati, New Age International publication, 2002.
2. Monochrome and color TV-R.R Gulati, New Age International publication, 2002.

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REFERENCES:

1. Color Television Theory and practice-S.P.Bali, TMH, 1994.
2. Television and Video engineering-A.M. Dhake, 2nd Edition, Tata McGraw Hill.
3. Basic television and video systems-B.Grob and C.E.Herndon, McGraw Hill, 1999.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04706) DIGITAL DESIGN THROUGH VERILOG HDL
(ELECTIVE – II)**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**INTRODUCTION TO VERILOG:**

Verilog as HDL, Levels of design description, concurrency, simulation and synthesis, functional verification, system tasks, programming language interface(PLI), module, simulation and synthesis tools, test benches.

LANGUAGE CONSTRUCTS AND CONVENTIONS:

Introduction, keywords, identifiers, whitespace characters, comments, numbers, strings, logic values, strengths, data types, scalars and vectors, parameters, memory, operators, system tasks, exercises.

UNIT II**GATE LEVEL MODELLING:**

Introduction, AND gate primitive, module structure, other gate primitives, illustrative examples, tristate gates, array of instances of primitives, additional examples, design of Flip flops with gate primitives, delays, strengths and contention resolution, net types, design of basic circuits, exercises.

UNIT III**BEHAVIORAL MODELLING:**

Introduction, operations and assignments, functional Bifurcation, initial construct, always construct, examples, assignments with delays, wait construct, multiple always blocks, designs at behavioral level, blocking and non-blocking assignments, the case statement, simulation flow, if and if else constructs, assign-De assign construct, repeat construct, FOR loop, the disable construct, While loop, Forever loop, parallel blocks, force-release construct, event.

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UNIT IV**MODELLING AT DATAFLOW LEVEL:**

Introduction, continuous assignment structures, delays and continuous assignments, assignment to vectors, operators.

SWITCH LEVEL MODELLING: Introduction, basic transistor switches, CMOS switch, Bidirectional gates, time delays with switch primitives, instantiations with strengths and delays, strength contention with trireg nets, exercises.

UNIT V**SYSTEM TASKS, FUNCTIONS AND COMPILER****DIRECTIVES:**

Introduction, parameters, path delays, module parameters, system tasks and functions, file –based tasks and Functions, Compiler Directives, Hierarchical Access, General Observations, exercises.

FUNCTIONS, TASKS, AND USER-DEFINED PRIMITIVES:

Introduction, Function, Tasks, User- Defined Primitives (UDP), FSM Design (Moore and Mealy Machines).

UNIT VI**DIGITAL DESIGN WITH SM CHARTS:**

State Machine Charts, Derivation of SM Charts, Realization of SM Charts, Implementation of the Dice Game, Alternative realizations for SM Charts using Microprogramming, Linked State Machines.

UNIT VII**DESIGNING WITH PROGRAMMABLE GATE ARRAYS AND COMPLEX PROGRAMMABLE LOGIC DEVICES:**

Xilinx 3000 Series FPGAs, Designing with FPGAs, Using a One-Hot State Assignment, Altera Complex Programmable Logic Devices (CPLDs), Altera FLEX 10K Series CPLDs.

UNIT VIII**VERILOG MODELS:**

Static RAM Memory, A simplified 486 Bus Model, Interfacing Memory to a Microprocessor Bus, UART Design, Design of Microcontroller CPU.

2009-10**TEXT BOOKS:**

1. Design through Verilog HDL – T.R. Padmanabhan and B. Bala Tripura Sundari, WSE, IEEE Press, 2004.
2. A Verilog Primer – J. Bhasker, BSP, 2003.

REFERENCES:

1. Fundamentals of Logic Design with Verilog – Stephen. Brown and Zvonko Vranesic, TMH, 2005.
2. Advanced Digital Design with Verilog HDL – Michael D. Ciletti, PHI, 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04707) MICROWAVE & OPTICAL COMMUNICATIONS
LAB**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
0	3	2

Minimum Twelve Experiments to be conducted:**Part – A (Any 7 Experiments):**

1. Reflex Klystron Characteristics.
2. Gunn Diode Characteristics.
3. Attenuation Measurement.
4. Directional Coupler Characteristics.
5. VSWR Measurement.
6. Impedance Measurement.
7. Waveguide parameters measurement.
8. Scattering parameters of Directional Coupler.
9. Scattering parameters of Magic Tee.

Part – B (Any 5 Experiments):

1. Characterization of LED.
2. Characterization of Laser Diode.
3. Intensity modulation of Laser output through an optical fiber.
4. Measurement of Data rate for Digital Optical link.
5. Measurement of NA.
6. Measurement of losses for Analog Optical link.
7. Radiation Pattern Measurement of Antennas (at least two antennas).

Equipment required for Laboratories:

1. Regulated Klystron Power Supply
2. VSWR Meter
3. Micro Ammeter
4. Multi meter
5. CRO
6. GUNN Power Supply, Pin Moderator

7. Reflex Klystron
8. Crystal Diodes
9. Micro wave components (Attenuation)
10. Frequency Meter
11. Slotted line carriage
12. Probe detector
13. wave guide shorts
14. Pyramidal Horn Antennas
15. Directional Coupler
16. E, H, Magic Tees
17. Circulators, Isolator
18. Matched Loads
19. Fiber Optic Analog Trainer based LED
20. Fiber Optic Analog Trainer based laser
21. Fiber Optic Digital Trainer
22. Fiber cables - (Plastic, Glass)
23. Antenna Training System with Tripod and Accessories

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04708) MICRO PROCESSORS & DSP LAB**

B.Tech IV-I Sem. (E.C.E.)

T	P	C
0	3	2

I. Microprocessor 8086 & Microcontroller 8051:**(Any four from 1 – 6, and 7, 8 are compulsory)**

1. Arithmetic operation – Multi byte Addition and Subtraction, Multiplication and Division – Signed and unsigned Arithmetic operation, ASCII – arithmetic operation.
2. Logic operations – Shift and rotate – Converting packed BCD to unpacked BCD, BCD to ASCII conversion.
3. By using string operation and Instruction prefix: Move Block, Reverse string, Sorting, Inserting, Deleting, Length of the string, String comparison.
4. Reading and Writing on a parallel port.
5. Timer in different modes.
6. Serial communication implementation.
7. 8259 – Interrupt Controller: Generate an interrupt using 8259 timer.
8. 8279 – Keyboard Display: Write a small program to display a string of characters.

II. DSP Processor: (Any six of the following)

1. To study the architecture of DSP chips – TMS 320C 5X/6X Instructions.
2. To verify linear convolution.
3. To verify the circular convolution.
4. To design FIR filter (LP/HP) using windowing technique
 - a) Using rectangular window
 - b) Using triangular window
 - c) Using Kaiser window
5. To Implement IIR filter (LP/HP) on DSP Processors.
6. N-point FFT algorithm.

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7. MATLAB program to find frequency response of analog LP/HP filters.
8. To compute power density spectrum of a sequence.

Equipment required for Laboratories:

1. 8086 μ P Kits
2. 8051 Micro Controller kits
3. Interfaces/peripheral subsystems
 - i) 8259 PIC
 - ii) 8279-KB/Display
 - iii) 8255 PPI
 - iv) 8251 USART
4. ADC Interface
5. DAC Interface
6. Traffic Controller Interface
7. Elevator Interface

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04801) CELLULAR & MOBILE COMMUNICATIONS**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I**CELLULAR MOBILE RADIO SYSTEMS:**

Introduction to Cellular Mobile system, performance criteria, uniqueness of mobile radio environment, operation of cellular systems, Hexagonal shaped cells, Analog and Digital Cellular systems.

UNIT II**ELEMENTS OF CELLULAR RADIO SYSTEM DESIGN:**

General description of the problem, concept of frequency channels, Co-channel Interference Reduction Factor, desired C/I from a normal case in a Omni directional Antenna system, Cell splitting, consideration of the components of cellular system.

UNIT III**INTERFERENCE:**

Introduction to Co-channel interference, real time co-channel interference, Co-channel measurement, design of Antenna system, Antenna parameters and their effects, diversity receiver, non-co-channel interference-different types.

UNIT IV**CELL COVERAGE FOR SIGNAL AND TRAFFIC:**

Signal reflections in flat and hilly terrain, effect of human made structures, phase difference between direct and reflected paths, constant standard deviation, straight line path loss slope, general formula for mobile propagation over water and flat open area, near and long distance propagation antenna height gain, form of a point to point model.

UNIT V**CELL SITE AND MOBILE ANTENNAS:**

Sum and difference patterns and their synthesis, Omni directional antennas, directional antennas for interference reduction, space diversity antennas, umbrella pattern antennas, minimum separation of cell site antennas, high gain antennas.

UNIT VI**FREQUENCY MANAGEMENT AND CHANNEL****ASSIGNMENT:**

Numbering and grouping, setup access and paging channels channel assignments to cell sites and mobile units, channel sharing and borrowing, sectorization, overlaid cells, non fixed channel assignment.

UNIT VII**HANDOFF:**

Handoff, dropped calls and cell splitting, types of handoff, handoff invitation, delaying handoff, forced handoff, mobile assigned handoff. Intersystem handoff, cell splitting, micro cells, vehicle locating methods, dropped call rates and their evaluation.

UNIT VIII**DIGITAL CELLULAR NETWORKS:**

GSM architecture, GSM channels, multiplex access scheme, TDMA, CDMA.

TEXT BOOKS:

1. Mobile cellular telecommunications-W .C. Y. Lee, Tata Mc-Graw Hill, 2nd Edition, 2006.
2. Wireless communications-Theodore. S. Rapport, Pearson Education, 2nd Edn., 2002.

REFERENCES:

1. Principles of Mobile communications-Gordon L. Stuber, Springer International 2nd Edition, 2007.
2. Wireless and Mobile Communications-Lee Mc Graw Hills, 3rd Edition, 2006.
3. Wireless communications and Networking-Jon W.Mark and Weihua Zhqung, PHI, 2005.
4. Wireless communication Technology-R.Blake, Thompson Asia Pvt.Ltd., 2004.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04802) DIGITAL IMAGE PROCESSING**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I**DIGITAL IMAGE FUNDAMENTALS:**

Image Sensing and Acquisition, Image Sampling & quantization, some basic Relationships between pixels. Mathematical tools used in digital image processing – array Vs matrix operations, linear Vs non linear operations, arithmetic operations, set and logical operations, spatial operations, vector and matrix operations, Probabilistic methods.

UNIT-II**IMAGE TRANSFORMS:**

2D-DFT and properties, Walsh Transform, Hadamard Transform, Discrete cosine Transform, Haar-Transform, Slant Transform, KL transform, comparison of different image transforms.

UNIT-III**IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN:**

Basic Intensity transformations functions, histogram Processing, fundamentals of Spatial Filtering, Smoothing Spatial filters, Sharpening spatial filters, Combining spatial enhancement methods.

UNIT-IV**IMAGE ENHANCEMENT IN FREQUENCY DOMAIN:**

Basics of filtering in frequency domain, additional characteristics of the frequency domain, correspondence between filtering in the spatial and frequency domains. Image smoothing using frequency domain filters, image sharpening using frequency domain filters – Gaussian High pass filters, Laplacian in the frequency domain, Homomorphic filtering.

UNIT-V**IMAGE DEGRADATION / RESTORATION:**

Noise models, Restoration in the presence of Noise only-spatial filtering, - mean, order- statistic and adaptive filters, Estimating the Degradation function, Inverse filtering, Weiner filtering, Constrained Least squares filtering.

UNIT-VI**IMAGE SEGMENTATION:**

Point, line and edge Detection, Thresholding, Region based segmentation, the use of motion in segmentation.

UNIT-VII**IMAGE COMPRESSION:**

Need for Image compression, Classification of Redundancy in Images, Image compression models, Classification of image compression schemes, Run length coding, arithmetic coding, Block truncation coding, Dictionary based compression, transform based compression, Image compression standards, Scalar quantization, vector quantization.

UNIT-VIII**COLOR IMAGE PROCESSING:**

Color models, pseudo color image processing, color transformations, Smoothing and sharpening, image segmentation based on color.

TEXT BOOKS:

1. Digital Image Processing-R. C .Gonzalez & R.E. Woods, Addison Wesley/Pearson education, 3rd Edition, 2010.
2. Digital Image processing– S jayaraman, S Esakkirajan, T Veerakumar, Tata McGraw Hill.

REFERENCES:

1. Digital Image processing using MATLAB-Rafael C. Gonzalez, Richard E woods and Steven L.Eddins, Tata McGraw Hill, 2010.
2. Fundamentals of Digital Image processing-A .K. Jain, PHI.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04803) SATELLITE COMMUNICATIONS
(ELECTIVE – III)**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT-I**INTRODUCTION:**

Origin of satellite communications, Historical background, basic concepts of satellite communications, frequency allocations for satellite services, applications, future trends of satellite communications.

UNIT-II**ORBITAL MECHANICS AND LAUNCHERS:**

Orbital Mechanics look angle determination, orbital perturbations, orbit determination, launches and launch vehicles, orbital effects in communication systems performance.

UNIT-III**SATELLITE SUBSYSTEMS:**

Attitude and orbital control system, Telemetry, Tracking, command and monitoring, power systems, communication subsystems, satellite antenna equipment reliability and space qualification.

UNIT-IV**SATELLITE LINK DESIGN:**

Basic transmission theory, system noise temperature and G/T ratio, design of down links, uplink design, design of satellite links for specified C/N, system design example.

UNIT-V**MULTIPLE ACCESS:**

Frequency division multiple access (FDMA) Intermodulation, calculation of C/N, Time Division multiple access (TDMA) frame structure, examples. Satellite switched TDMA onboard processing,

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DAMA, code division multiple access (CDMA), spread spectrum transmission and reception.

UNIT-VI

EARTH STATION TECHNOLOGY:

Introduction, transmitters, receivers, Antennas, tracking systems, terrestrial interface, primary power test methods.

UNIT-VII

LOW EARTH ORBIT AND GEO-STATIONARY SATELLITE SYSTEMS:

Orbit consideration, coverage and frequency considerations, delay and throughput considerations, system considerations, operational NGSO constellation designs.

UNIT-VIII

SATELLITE NAVIGATION & THE GLOBAL POSITIONING SYSTEM:

Radio and satellite navigation, GPS position location principles, GPS receivers and codes, satellite signal acquisition, GPS navigation message, GPS signal levels, GPS receiver operation, GPS C/A code accuracy, differential GPS.

TEXT BOOKS:

1. Satellite communications-Timothi Pratt, Charles Bostian and Jeremy Allnutt, WSE, Wiley publications, 2nd Edition, 2003.
2. Satellite communications Engineering-Wilbur L.Prichard, Robert A. Nelson & Henry G.Suyderhoud, 2nd Edition, Pearson Publications, 2003.

REFERENCES:

1. Satellite communications: Design principles-M. Richharia, BS publications, 2nd Edition, 2003.
2. Satellite communications-D.C.Agarwal, Khanna publications, 5th Ed.
3. Fundamentals of Satellite communications-K.N.Raja rao, PHI, 2004.
4. Satellite communications-Dennis Roddy, McGraw Hill, 2nd Edition, 1996.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04804) SPREAD SPECTRUM COMMUNICATIONS
(ELECTIVE – III)**

B.Tech IV-II Sem. (E.C.E.)	T	P	C
	4	0	4

UNIT I**FUNDAMENTALS OF SPREAD SPECTRUM:**

General concepts, Direct sequence (DS), Pseudo Noise (PN), Frequency Hopping, Time Hopping, Comparison of Modulation methods, Hybrid Spread spectrum systems, Chirp spread spectrum, Baseband modulation techniques.

UNIT II**ANALYSIS OF DIRECT SEQUENCE SPREAD SPECTRUM SYSTEMS:**

Properties of PN sequences, Classes of periodic sequences, Properties of m sequences, Partial Co-relation, PN signal from PN sequences, Partial co – relation of PN signals, The PN Signal, De-spreading the PN signal, Interference rejection, Output signal to noise ratio, Antijam characteristics, Interception, Energy bandwidth efficiency.

UNIT III**ANALYSIS OF AVOIDANCE – TYPE SPREAD SPECTRUM SYSTEMS:**

The frequency hopped signal, Interference rejection in a frequency hopping receiver, the time hopped signal.

UNIT IV**GENERATION OF SPREAD SPECTRUM SIGNALS:**

Shift register sequence generators, Discrete frequency synthesizers, SAW device PN generators, Charge coupled devices, Digital tapped delay lines.

UNIT V**DETECTION OF SPREAD SPECTRUM SIGNALS - TRACKING:**

Coherent direct sequence receivers, other method of carrier tracking, Delay lock loop analysis, Tau – Dither loop, Coherent carrier tracking, Non coherent frequency hop receiver.

UNIT VI**DETECTION OF SPREAD SPECTRUM SIGNALS - ACQUISITION:**

Acquisition of spread spectrum signals, Acquisition cell by cell searching, Reduction of acquisition time, Acquisition with matched filters, Matched filters for PN sequences, Matched filters for frequency hopped signals, Matched filters with acquisition - aiding waveform.

UNIT VII**APPLICATION OF SPREAD SPECTRUM TO COMMUNICATIONS:**

General capabilities of spread spectrum, Multiple access considerations, Energy and bandwidth efficiency in multiple access, Selective calling and Identification, Antijam considerations, Error correction coding, Intercept consideration (AI), Miscellaneous considerations, Examples of spread spectrum systems.

UNIT VIII**CODE DIVISION MULTIPLE ACCESS DIGITAL CELLULAR SYSTEMS:**

Introduction, Cellular radio concept, CDMA Digital cellular systems, Specific examples of CDMA digital cellular systems.

TEXT BOOKS:

1. George. R. Cooper and Clare D. McGillem, “Modren Communications and Spread Spectrum”, McGraw hill Book Company.
2. Roger L. Peterson, Rodger E. Ziemer & David E. Borth, “Introduction to Spread Spectrum Communications”, Prentice Hall 1995.

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REFERENCES:

1. Dr. Kamilo Feher, “Wireless Digital Communications – Modulation & Spread Spectrum Applications”, PHI, 1999.
2. Upena Dalal, “Wireless Communication”, Oxford Higher Education, 2009.
3. Andrea Goldsmith “Wireless Communications”, Cambridge University Press, 2005.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A05709) INFORMATION SECURITY
(Common to ECE, ECM)
(ELECTIVE – III)**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I

Is There A Security Problem in Computing: What Does Security Mean?, Attacks, The Meaning Of Computer Security, Computer Criminals, Methods of Defense, Terminology and Background, Substitution Ciphers, Transpositions(Permutations), Making good Encryption Algorithm, The Data Encryption Standard.

UNIT II

Program Security: Secure Programs, Non-Malicious Program Errors, Viruses and Other Malicious Code, Targeted Malicious Code.

UNIT III

Public-Key Cryptography and RSA, Key Management; Other public key Cryptosystems, Message Authentication and Hash Functions: Authentication Requirements, Authentication Functions, Message Authentication Codes, Hash Functions, Security Hash Functions and MACs, Hash and MAC Algorithms: Secure Hash Algorithm, Whirlpool.

UNIT IV

Digital Signatures and Authentication Protocols: Digital Signatures, Authentication Protocols.

UNIT V

Authentication Applications: Kerberos, Electronic Mail Security: Pretty Good Privacy, S/MIME.

2009-10

UNIT VI

IP Security: IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations, Key Management.

UNIT VII

Web Security: Web Security Considerations, Secure Socket Layer and Transport Layer Security, Secure Electronic Transaction.

UNIT VIII

Intruders: Intruders, Intrusion Detection, Password Management, Firewalls: Firewall Design and Principles, Trusted Systems.

TEXT BOOKS:

1. Security in Computing, Charles P. Pfleeger, Shari Lawrence Pfleeger, Deven Shah, Pearson Education.
2. Cryptography and Network Security, William Stallings Fourth Edition, Pearson Education.

REFERENCES:

1. Cryptography and Information Security, V.K.Pachghare, PHI.
2. Information Security -Theory and Practice, Dhiren R. Patel, PHI.
3. Introduction to Cryptography, Buchmann, Springer.
4. Principles and Practices of Information Security, Michael E. Whitman and Herbert J. Mattord, Cengage Learning.
5. Information Systems Security, Godbole, Wiley Student Edition.

2009-10

**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04805) ADAPTIVE FILTER THEORY
(ELECTIVE – IV)**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT – I**INTRODUCTION**

The filtering problem, Adaptive filters, Linear Filter Structures, Approaches to the development of Linear Filtering algorithms, Real and Complex forms of Adaptive filters, Applications.

UNIT – II**STATIONARY PROCESSES AND MODELS**

Partial characterization of a discrete-time stochastic process, Mean ergodic theorem, Correlation matrix and its properties, Stochastic models, Wold decomposition, Yule-Walker Equations, Complex Gaussian processes.

UNIT – III**EIGENANALYSIS**

The Eigen value problem, Properties of Eigen values and eigenvectors, Low-rank modeling, Eigen filters, Eigen value computations.

UNIT – IV**WIENER FILTERS**

Linear optimum filtering, Principle of orthogonality, Minimum mean squared error, Wiener-Hop equations, Error performance surface, Numerical example, Channel equalization, Linearly constrained minimum variance filter.

UNIT – V**LINEAR PREDICTION**

Forward, and backward linear prediction, Properties of Prediction error filters, Autoregressive Modeling of stationary stochastic process, Cholesky factorization, Lattice Predictors.

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UNIT – VI**KALMAN FILTERS**

Recursive Minimum mean square estimation for scalar random variables, Statement of the Kalman filtering problem, Estimation of the state using the innovations process, Filtering, initial conditions, summary of the Kalman filter.

UNIT – VII**LINEAR ADAPTIVE FILTERING - I**

Steepest Descent algorithm, example, characterization of the AR process, Least Mean Squared (LMS) algorithm, Examples, stability and performance analysis of the LMS algorithm, Summary of the LMS algorithm.

UNIT – VIII**LINEAR ADAPTIVE FILTERING – II**

Method of Least Squares, Statement of the linear Least Squares-Estimation Problem, Data windowing, Principle of orthogonality, Singular value decomposition, Recursive Least Squares (RLS) algorithm, the matrix inverse lemma, exponentially weighted recursive least squares algorithm, time update of the tap weight vector, summary of RLS algorithm, examples.

TEXT BOOKS:

1. Simon Haykin, “Adaptive Filter Theory,” Prentice Hall, New Jersey, 3rd Edition, 1996.
2. Bernard Widrow, and Samuel D. Stearns, “Adaptive Signal Processing,” Prentice Hall, New Jersey, 1985.

REFERENCES:

1. John R. Treichler, C. R. Johnson, Jr., and M. G. Larimore, “Theory and Design of Adaptive Filters,” PHI Learning Pvt. Ltd, New Delhi, 2009.
2. Alexander D. Poularikas, Zayed M. Ramadan, “Adaptive Filtering Primer with MATLAB,” Taylor and Francis A CRC Press, N Y, 2006.
3. Antoniou, “Adaptive Filter Design,” New York, NY: John Wiley and Sons, Inc., 1987.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04806) WIRELESS COMMUNICATIONS & NETWORKS
(Common to ECE, ECM)
(ELECTIVE – IV)**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I

MULTIPLE ACCESS TECHNIQUES FOR WIRELESS COMMUNICATION: Introduction, FDMA, TDMA, Spread Spectrum, Multiple Access, SDMA, Packet radio, Packet radio protocols, CSMA protocols, Reservation protocols.

UNIT II**INTRODUCTION TO WIRELESS NETWORKING:**

Introduction, Difference between wireless and fixed telephone networks, Development of wireless networks, Traffic routing in wireless networks.

UNIT III**WIRELESS DATA SERVICES:**

CDPD, ARDIS, RMD, Common channel signaling, ISDN, BISDN and ATM, SS7, SS7 user part, signaling traffic in SS7.

UNIT IV**MOBILE IP AND WIRELESS ACCESS PROTOCOL:**

Mobile IP Operation of mobile IP, Co-located address, Registration, Tunneling, WAP Architecture, overview, WML scripts, WAP service, WAP session protocol, wireless transaction, Wireless datagram protocol.

UNIT V**WIRELESS LAN TECHNOLOGY:**

Infrared LANs, Spread spectrum LANs, Narrow band microwave LANs, IEEE 802 protocol Architecture, IEEE802 architecture and services, 802.11 medium access control, 802.11 physical layer.

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UNIT VI**BLUE TOOTH:**

Overview, Radio specification, Base band specification, Links manager specification, Logical link control and adaptation protocol. Introduction to WLL Technology.

UNIT VII**MOBILE DATA NETWORKS:**

Introduction, Data oriented CDPD Network, GPRS and higher data rates, Short messaging service in GSM, Mobile application protocol.

UNIT VIII**WIRELESS ATM & HIPER LAN:**

Introduction, Wireless ATM, HIPERLAN, Adhoc Networking and WPAN.

TEXT BOOKS:

1. Wireless Communications, Principles, Practice – Theodore S. Rappaport, PHI, 2nd Ed., 2002.
2. Wireless Communication and Networking – William Stallings, PHI, 2003.

REFERENCES:

1. Wireless Digital Communications – Kamilo Feher, PHI, 1999.
2. Principles of Wireless Networks – Kaveh Pah Laven and P. Krishna Murthy, Pearson Education, 2002.
3. Wireless Communications – Andreaws F. Molisch, Wiley India, 2006.
4. Introduction to Wireless and Mobile Systems – Dharma Prakash Agarwal, Qing-An Zeng, Thomson 2nd Edition, 2006.

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**JAWAHARLAL NEHRU
TECHNOLOGICAL UNIVERSITY ANANTAPUR**

**Electronics and Communication Engineering
(9A04807) DATA COMMUNICATIONS
(Common to E Con E, ECE)
(ELECTIVE – IV)**

B.Tech IV-II Sem. (E.C.E.)

T	P	C
4	0	4

UNIT I

INTRODUCTION TO DATA COMMUNICATIONS AND NETWORKING:

Standards Organizations for Data Communications, Layered Network Architecture, Open Systems Interconnection, Data Communications Circuits, Serial and parallel Data Transmission, Data communications Circuit Arrangements.

SIGNALS, NOISE, MODULATION, AND DEMODULATION:

Signal Analysis, Electrical Noise and Signal-to-Noise Ratio, Analog Modulation Systems, Information Capacity, Bits, Bit Rate, Baud, and M-ary Encoding, Digital Modulation.

UNIT II

METALLIC CABLE TRANSMISSION MEDIA:

Metallic Transmission Lines, Transverse Electromagnetic Waves, Characteristics of Electromagnetic Waves, Transmission Line Classifications, Metallic Transmission Line Types, Metallic Transmission Line Equivalent Circuit, Wave Propagation on Metallic Transmission Lines, Metallic Transmission Line Losses.

OPTICAL FIBER TRANSMISSION MEDIA:

Advantages of Optical Fiber Cables, Disadvantages of Optical Fiber Cables, Electromagnetic spectrum, Optical Fiber Communications System Block Diagram, Optical Fiber construction, The Physics of Light, Velocity of Propagation, Propagation of Light Through an Optical fiber Cable, Optical Fiber Modes and Classifications, Optical Fiber Comparison, Losses in Optical Fiber Cables, Light sources, Light Detectors, Lasers.

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UNIT III**DIGITAL TRANSMISSION:**

Pulse Modulation, Pulse code Modulation, Dynamic Range, Signal Voltage –to-Quantization Noise Voltage Ration, Linear Versus Nonlinear PCM Codes, Companding, PCM Line Speed, Delta Modulation PCM and Differential PCM.

MULTIPLEXING AND T CARRIERS:

Time- Division Multiplexing, T1 Digital Carrier System, North American Digital Multiplexing Hierarchy, Digital Line Encoding, T Carrier systems, European Time- Division Multiplexing, Statistical Time – Division Multiplexing, Frame Synchronization, Frequency-Division Multiplexing, Wavelength- Division Multiplexing, Synchronous Optical Network.

UNIT IV**WIRLESS COMMUNICATIONS SYSTEMS:**

Electromagnetic Polarization, Rays and Wavefronts, Electromagnetic Radiation, Spherical Wavefront and the Inverse Square Law, wave Attenuation and Absorption, Optical Properties of Radio Waves, Terrestrial Propagation of Electromagnetic Waves, Skip Distance, Free-Space Path Loss, Microwave Communications Systems, Satellite Communications Systems.

UNIT V**TELEPHONE INSTRUMENTS AND SIGNALS:**

The Subscriber Loop, Standard Telephone Set, Basic Telephone Call Procedures, Call Progress Tones and Signals, Cordless Telephones, Caller ID, Electronic Telephones, Paging systems.

THE TELEPHONE CIRCUIT:

The Local Subscriber Loop, Telephone Message- Channel Noise and Noise Weighting, Units of Powers Measurement, Transmission Parameters and Private-Line Circuits, Voice-Frequency Circuit Arrangements, Crosstalk.

UNIT VI**CELLULAR TELEPHONE SYSTEMS:**

Concepts – Frequency reuse- Cell splitting – Network components – Call Processing - First- Generation Analog Cellular Telephone,

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Personal Communications system, Second-Generation Cellular Telephone Systems, N-AMPS, Digital Cellular Telephone, Global system for Mobile Communications.

UNIT VII**DATA COMMUNICATIONS CODES, ERROR CONTROL, AND DATA FORMATS:**

Data Communications Character Codes, Bar Codes, Error Control, Error Detection, Error Correction, Character Synchronization.

DATA COMMUNICATIONS EQUIPMENT:

Digital Service Unit and Channel Service Unit, Voice- Band Data Communication Modems, Bell Systems- Compatible Voice- Band Modems, Voice- Band Modem Block Diagram, Voice- Band Modem Classifications, Asynchronous Voice-Band Modems, Synchronous Voice-Band Modems, Modem Synchronization, ITU-T Voice- Band Modem Specifications, 56K Modems, Modem Control: The AT Command Set, Cable Modems, Probability of Error and Bit Error Rate.

UNIT VIII**DATA –LINK PROTOCOLS:**

Data –Link Protocol Functions, Character –and Bit- Oriented Protocols, Data Transmission Modes, Asynchronous Data – Link Protocols, Synchronous Data – Link Protocols, Synchronous Data – Link Control, High – Level Data – Link Control.

TEXT BOOKS:

1. Introduction to Data Communications and Networking, Wayne Tomasi, Pearson Education.

REFERENCES

1. Data Communications and Networking, Behrouz A Forouzan, 4th Edition, TMH.
2. Computer Communications and Networking Technologies, Gallow, 2nd edition, Thomson.

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3. Computer Networking and Internet, Fred Halsll, Lingana Gouda Kulkarni, 5th Edition, Pearson Education.